



Ultra-Low-Power Neuromorphic AI for Epileptic Seizure Detection

A Spatio-Temporal CNN to Spiking Neural Network Pipeline
Deployed on BrainChip AKD1000 Silicon

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Abstract

Continuous, wearable seizure monitoring offers transformative potential for ambulatory epilepsy care, but practical deployment is fundamentally constrained by strict power budgets rather than algorithmic accuracy in isolation. This dissertation reports an end-to-end pipeline – a quantised (w4a4) spatio-temporal convolutional neural network, converted to a spiking neural network and deployed on real BrainChip AKD1000 v1 silicon – and evaluates it against the CHB-MIT paediatric scalp EEG dataset (Goldberger et al., 2000) with an explicit emphasis on clinical and chronological honesty throughout.

A full 15-fold leave-one-patient-out (LOPO) cross-validation gives an all-folds mean event sensitivity of 0.586 (± 0.445), a figure shown to be inflated by a 47% collapse-artefact; the corrected, collapse-diagnostic figure is 0.470 (± 0.437). The shape of this result – markedly bimodal rather than smoothly distributed – is presented as evidence for genuine inter-patient physiological heterogeneity rather than a simple data-volume shortfall.

To address this heterogeneity, the dissertation isolates a single variable – whether a patient’s own data is present in an otherwise near-constant-sized training pool – finding a real or artefact-clarifying benefit in five of seven patients. A smaller, secondary evaluation of per-patient fine-tuning contributes a pool-size finding (collapse-pass rate improving from 37.5% to 87.5% with a larger pool) and uses split-conformal prediction to correct an optimistic, threshold-selected false-positive-rate (FP/hr) figure.

On real AKD1000 v1 hardware, measured energy cost is approximately 0.906 μ J per inference (chip-isolated 1mW active power), alongside a separately measured, and separately scoped, system-level incremental power draw of 5.63W – the two figures are reported side by side rather than one validating the other, since they describe genuinely different quantities.

Throughout, this dissertation adopts a triage-device framing, analogous to D-dimer testing in the diagnosis of pulmonary embolism: a low-power, sensitivity-prioritised first-tier screen intended to flag events for closer attention, not a stand-alone replacement for unconstrained-compute clinical systems. Presented alongside its headline results are the project’s own caught and corrected methodological errors, treated as evidence of rigour rather than omitted. This work is offered as a first, deliberately modest step toward a genuinely deployable, low-power seizure-screening capability – one intended, in time and with further validation, to be of real benefit to people living with epilepsy, rather than as a final claim to clinical readiness.

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Acronyms and Notation

Acronyms

Acronym	Meaning
ANN	Artificial Neural Network
AUPRC	Area Under the Precision-Recall Curve
BN	Batch Normalisation
C1	Frozen shared-pool training regime (chb01/02/05)
C2	Per-patient fine-tuning training regime
CHB-MIT	Children’s Hospital Boston – MIT (scalp EEG dataset)
CNN	Convolutional Neural Network
CORAL	Correlation Alignment (domain-adaptation method)
DANN	Domain-Adversarial Neural Network
EEG	Electroencephalography / Electroencephalogram
FP/hr	False Positives per Hour
FPR	False Positive Rate
LIF	Leaky Integrate-and-Fire (spiking neuron model)
LOPO	Leave-One-Patient-Out (cross-validation)
NPU	Neural Processing Unit (AKD1000 hardware unit)
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
SMOTE	Synthetic Minority Over-sampling Technique
SNN	Spiking Neural Network
SSL	Self-Supervised Learning (pretraining)
STFT	Short-Time Fourier Transform
w4a4	4-bit weights, 4-bit activations (quantisation scheme)

Notation

Symbol	Meaning
$V(t)$	LIF neuron membrane potential at time t
τ	LIF membrane time constant
R	LIF neuron input resistance
$I(t)$	Incoming current to a LIF neuron at time t
V_{th}	LIF spike threshold potential
V_{reset}	LIF post-spike reset potential
α	Conformal prediction tolerance level (Section 3.4)
chbXX	CHB-MIT patient identifier (e.g. chb10, chb13)

Chapter 1

Introduction

1.1 Motivation

Epilepsy affects approximately 50 million people worldwide (World Health Organization, 2024), and a substantial minority are still experiencing unpredictable seizures despite pharmacological management. Continuous, ambulatory monitoring offers the possibility of timely intervention, caregiver alerting, and long-term seizure quantification for treatment planning, but the practical barrier to any such system is power. A device that must be worn continuously, for weeks or months at a time, on a battery small enough to be tolerable, cannot draw on the multi-watt power budgets that conventional convolutional neural network inference assumes. This dissertation is concerned with a narrower and more concrete question than “can a neural network detect seizures from scalp EEG” – a question that has been answered affirmatively many times over – and asks instead whether that detection capability survives being compressed onto neuromorphic silicon operating at microwatt-scale power, and what is honestly lost or gained in doing so.

Neuromorphic hardware, and the BrainChip AKD1000 processor specifically, offers a route to this power budget by replacing dense, clocked matrix multiplication with sparse, event-driven spiking computation. The cost of this route is a set of hard architectural constraints – low-bitwidth quantisation, restricted layer types, no native support for some operations taken for granted on conventional accelerators – that shape every design decision in this project, from the network architecture (Chapter 3) to what forms of temporal context are even representable on the target silicon. This dissertation reports what happens when a real, physical AKD1000 chip, not a simulator, is asked to perform this task end to end.

A further commitment, developed fully in Chapter 3 but stated here because it shapes every result in this report: all data splits in this project are chronological, not randomly stratified. An earlier portion of each patient’s recording is used for training and an earlier-still portion for calibration, with a later, strictly held-out portion reserved for testing, so

that no information from a point in time after a given test window can ever have influenced the model’s behaviour at that window – a guarantee a real, prospectively-deployed device would also have to satisfy, and one a randomly stratified split cannot make.

It is worth stating plainly, at the outset, what this project does and does not claim to be. It is not a finished clinical product, nor does it claim readiness for deployment at scale. It is a first, deliberately modest step: a demonstration that the pipeline is technically feasible end to end, on real hardware, evaluated with the same rigour that a larger validation effort would require, so that a genuine, staged path toward wider deployment – more patients, more sites, a miniaturised and regulatory-cleared form factor – has an honest foundation to build from rather than an inflated one.

1.2 The Triage-Device Framing

It would be easy, and misleading, to present this system as an attempt to outperform state-of-the-art seizure detectors that assume unconstrained compute. It does not, and is not designed to. The more accurate clinical analogy, adopted throughout this dissertation, is the D-dimer blood test used in the emergency diagnosis of pulmonary embolism (PE).

D-dimer testing is not a diagnosis of PE. It is a cheap, fast, low-risk first-line screen with deliberately high sensitivity and correspondingly modest specificity: a negative result is used to rule PE out with confidence, while a positive result does not confirm PE but instead triggers a more expensive, more specific confirmatory investigation, typically CT pulmonary angiography. The clinical value of the test lies precisely in its asymmetry – it is acceptable, even desirable, for the test to over-flag, provided it essentially never under-flags a genuine case, because the cost of a false positive (an unnecessary confirmatory scan) is far smaller than the cost of a missed embolism.

This project adopts the same asymmetry deliberately. Event sensitivity – not missing a genuine seizure – is treated throughout as the primary metric; this was a design decision I made on the grounds that a low-power triage-tier device should be judged first on what it fails to catch. False-positive rate (FP/hr) is never omitted alongside sensitivity, nor is it treated as a secondary or cosmetic concern, but it is not the metric this system is optimised to minimise at the expense of missed events. A microwatt-class device sitting on a patient’s scalp is not positioned to replace a full clinical monitoring suite; it is positioned to run continuously, cheaply, and reliably enough to flag something worth a closer, more expensive “look happened here” – always-on first tier, not a stand-alone diagnostic authority. Where this dissertation’s results fall short of unconstrained-compute comparators (Chapter 4), this framing is the honest context for that gap, not an excuse for it: the two systems are answering different clinical questions under different resource constraints, and the comparison is presented with that caveat stated plainly rather than glossed over.

1.3 Research Questions

This dissertation is organised around four research questions.

RQ1. Can a quantised (w4a4) spatio-temporal convolutional neural network, converted to a spiking neural network via ANN-to-SNN conversion, and deployed on real BrainChip AKD1000 v1 silicon, achieve clinically meaningful per-patient sensitivity at microjoule-level energy cost per inference?

RQ2. How far does a shared, cross-patient representation generalise to entirely unseen patients under a leave-one-patient-out (LOPO) evaluation, and what does the *shape* of that result – not merely its mean – reveal about heterogeneity in ictal presentation across patients?

RQ3. Does including a patient’s own data measurably improve that patient’s detection outcome, when this is isolated as the single variable under test – and separately, does the narrower technique of per-patient fine-tuning on top of a frozen shared model offer any further gain, or introduce sensitivities (to pool size, to training seed) worth disclosing in their own right?

RQ4. Where a technique’s headline figure depends on how its operating threshold was chosen, does that figure survive a stricter, more statistically grounded re-calibration, or was the original figure – however honestly obtained at the time – optimistic in a way that needs to be disclosed and corrected?

RQ4 exists because this project’s own evaluation practice was, at one stage, itself in need of exactly this kind of scrutiny (Section 4.3), and the resolution of that question is treated as a report finding in its own right.

1.4 What This Dissertation Covers

This dissertation reports, in full:

1. An end-to-end pipeline running from raw scalp EEG through a quantised CNN, ANN-to-SNN conversion, and deployment on physical AKD1000 v1 silicon, complete with measured, hardware-verified energy and latency figures.
2. A full 15-fold leave-one-patient-out cross-validation of cross-patient generalisation, accompanied by the bimodal distribution analysis that points to genuine inter-patient physiological heterogeneity rather than a simple data-volume shortfall.
3. A formal collapse diagnostic (gating on window specificity, predicted block length, and positive-window fraction) to systematically distinguish true seizure discrimination from over-firing and under-firing artefacts.

4. Primary personalisation evidence from a controlled `pool6-minus-X` versus `pool7` experiment, isolating target-patient data inclusion as a single variable across seven patients.
5. A secondary evaluation of per-patient fine-tuning, quantifying the impact of background pool size on collapse-pass rates alongside a split-conformal-prediction calibration to correct optimistic, threshold-selected false-positive rates (FP/hr).
6. Documented, mechanism-level negative results, including four independent domain-adaptation and feature-engineering candidates (DANN, CORAL, SSL pretraining, and Candidate G) converging on the same patient’s structural failure.

What any of this amounts to, and whether it constitutes a meaningful advance, is left for the reader to judge against the evidence presented, rather than asserted here in advance.

1.5 Dissertation Structure

Chapter 2 (Background and Related Work) Reviews the clinical and technical background: the CHB-MIT dataset and standard evaluation conventions in automated seizure detection, the AKD1000’s neuromorphic architecture and its hard constraints, the ANN-to-SNN conversion toolchain, and the two comparator systems –Shoeb (2009) and Ali, Angelova, and Karmakar (2024) –against which this project’s per-patient and cross-patient results are respectively judged.

Chapter 3 (Methodology) Describes dataset preparation, model architecture, training regimes (the frozen shared pool, an isolated own-data- inclusion probe, per-patient fine-tuning, and a separate data-efficiency probe), and the evaluation protocol, including the distinction between full-recording LOPO evaluation and the smaller chronological test-slice used elsewhere in the pipeline.

Chapter 4 (Results) Reports the LOPO cross-patient generalisation result; the dissertation’s primary personalisation evidence, an isolated comparison showing the effect of including a patient’s own data in an otherwise-fixed training pool; a smaller, secondary treatment of per-patient fine-tuning framed around pool-size and training-seed effects, including the conformal-calibrated correction to an earlier optimistic figure; the cross-candidate convergence finding; and the hardware deployment figures – throughout stating any raw or optimistic figure in full before the corrected figure that supersedes it.

Chapter 5 (Discussion) Returns to the triage-device framing with concrete numbers in hand, discusses the physiological-heterogeneity account of cross-patient failure,

and treats the project's own caught methodological errors as a connected narrative and a contribution rather than a list of embarrassments.

Chapter 6 (Conclusion) Summarises the answers to RQ1–RQ4 and outlines future work.

Chapter 2

Background and Related Work

2.1 The Biology of Epileptic Seizures

Before discussing how seizures are detected computationally, it is worth being explicit about what a seizure actually is physiologically, since the detection problem this dissertation addresses is a direct consequence of it. A seizure is a transient episode of abnormally excessive or synchronous neuronal electrical activity in the brain: a large population of cortical neurons that would ordinarily fire in a loosely correlated, low-amplitude fashion instead begins firing in a highly synchronised, high-amplitude, rhythmic pattern, typically triggered by a local disruption to the normal balance between excitatory and inhibitory signalling in a region of cortex (World Health Organization, 2024). This abnormal synchrony can remain localised (a focal seizure) or spread across both hemispheres (a generalised seizure), and clinicians conventionally divide a seizure’s timeline into the *preictal* period beforehand, the *ictal* period during the seizure itself, and the *postictal* period of recovery afterward – terminology used throughout this dissertation, particularly in Chapter 4, where “ictal morphology” refers to the characteristic shape this synchronised electrical activity takes in a given patient’s recording.

Electroencephalography (EEG) is able to detect this abnormal synchrony non-invasively because scalp electrodes measure the summed electrical field potential generated by large populations of cortical neurons firing together; a small number of neurons firing independently produces a signal too weak and too spatially cancelled-out to register at the scalp, but the highly synchronised firing of a seizure produces a comparatively large, coherent voltage deflection that does register, which is precisely why EEG – rather than, say, single-neuron recording – is the practical tool of choice for this problem at scale. This also explains why seizure detection is fundamentally a pattern-recognition problem over time-series waveform morphology rather than a simple amplitude-threshold problem: a seizure is identifiable by its characteristic rhythmic, evolving waveform shape, not merely by being “louder” than the surrounding background EEG, and a substantial part of what

a detection model – whether a hand-crafted-feature classifier or a learned convolutional network, as in this project – must actually learn is this shape.

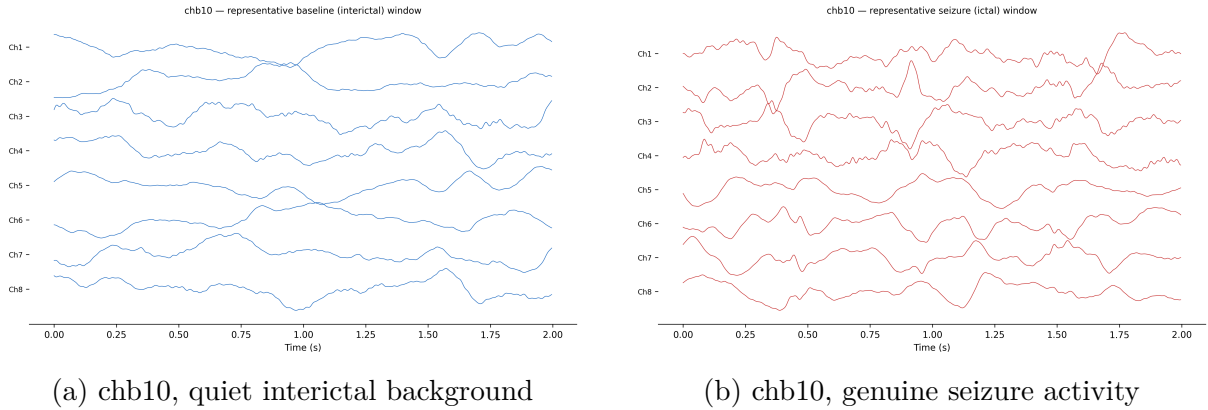


Figure 2.1: Raw EEG waveform comparison for patient chb10: quiet interictal background alongside genuine seizure activity, same channel and comparable time window.

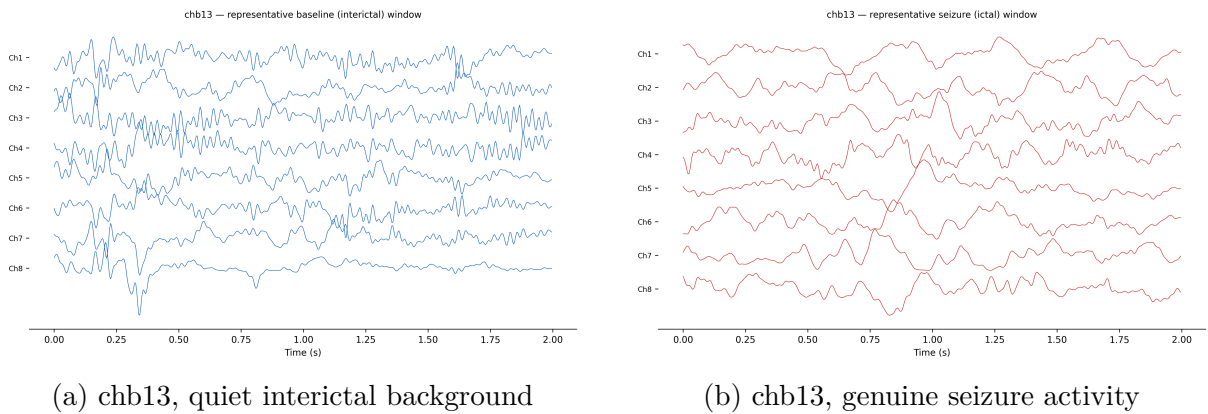


Figure 2.2: Raw EEG waveform comparison for patient chb13: quiet interictal background alongside genuine seizure activity, same channel and comparable time window.

Figures 2.1 and 2.2 together illustrate the waveform-morphology difference a detection model must learn to distinguish, and are shown for two different patients deliberately: comparing the two makes visible, even before any results are presented, that both the background EEG and the seizure waveform itself look noticeably different from one patient to the next – a first, purely visual hint at the inter-patient heterogeneity this dissertation returns to repeatedly from Chapter 4 onward.

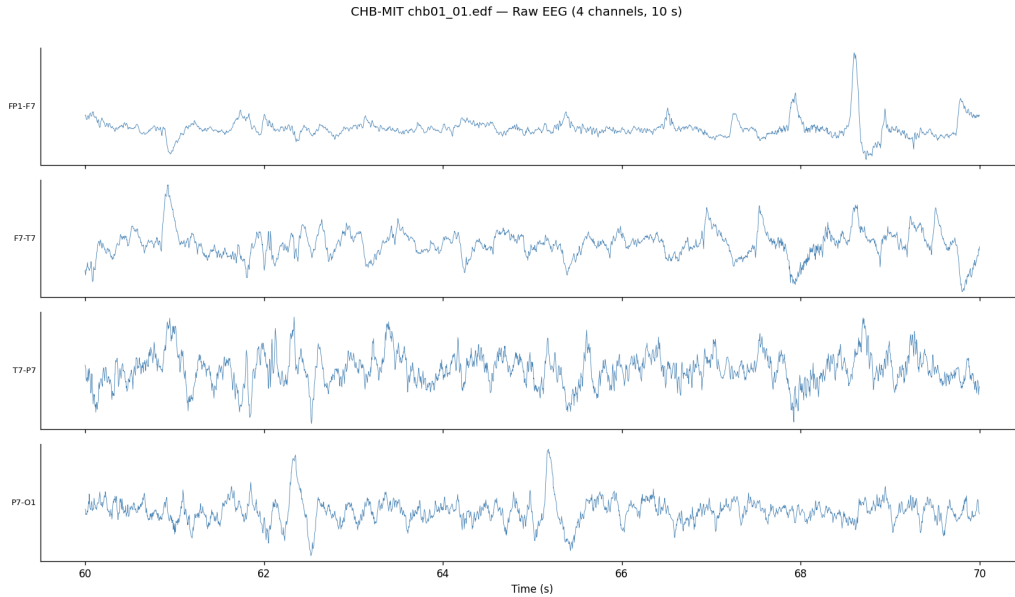


Figure 2.3: Ten seconds of real raw scalp EEG across four channels from `chb01_01.edf`, illustrating the multi-channel structure of the raw input signal before any windowing or labelling.

2.2 Epileptic Seizure Detection from Scalp EEG

Automated seizure detection from scalp electroencephalography (EEG) is a long-studied problem, and the CHB-MIT paediatric dataset (Goldberger et al., 2000) – continuous, long-duration recordings from young patients with intractable epilepsy, collected at Boston Children’s Hospital – has become one of its standard public benchmarks. The dataset’s realism is also its central difficulty: seizures are rare relative to interictal (non-seizure) background, recordings run for many hours per patient, and patients differ substantially from one another in electrode montage coverage, seizure semiology, and background EEG morphology.

Two evaluation conventions recur throughout the literature and are both used, for different purposes, in this dissertation. *Window-level* evaluation scores every fixed-length segment of EEG independently, as an ordinary binary classification problem; it is convenient for training but does not directly correspond to what a clinician or patient cares about, since a single seizure event may span many consecutive windows, none of which need to be individually correct for the event as a whole to be usefully flagged. *Event-level* evaluation instead asks whether each genuine seizure event was flagged at all (event sensitivity), and how often the system fires outside of any genuine event, expressed as a rate per hour of recording (FP/hr) rather than a raw count, so that patients with different total recording durations remain comparable. This dissertation treats event-level sensitivity and FP/hr as its primary reporting unit throughout, for the clinical reasons set out in Section 3.4, while retaining window-level metrics as a diagnostic tool – most

importantly, window-level specificity as the basis of a *collapse diagnostic* used throughout this dissertation. In brief, since it is used repeatedly from this point onward: a model that fires on almost every window it sees will register a high event sensitivity simply because it never stays silent through a genuine seizure, which is an artefact of near-permanent triggering rather than genuine discrimination; the collapse diagnostic is a gate, based on window-level specificity and predicted-block length, that catches this artefact before a sensitivity figure is reported as genuine. Its exact numerical criteria are given in full in Section 3.4.

2.3 Neuromorphic Computing and the BrainChip AKD1000

Conventional deep learning inference performs a multiply-accumulate operation at every neuron, at every clock tick, regardless of whether the underlying input has changed – a computational pattern well suited to GPU batching but poorly suited to a power budget measured in microwatts. Neuromorphic computing takes its name and its founding motivation from the alternative the biological brain demonstrates: sparse, event-driven computation, in which a neuron performs work only when it receives an incoming spike, and most neurons are silent at any given moment (Mead, 1990). Schuman et al. (2017) give a broad survey of how this principle has been implemented across a range of neuromorphic hardware platforms, of which BrainChip’s AKD1000 is one commercial example.

The AKD1000 v1 implements this event-driven principle in silicon via an array of Neural Processing Units (NPU) connected by an event-driven on-chip fabric, with synaptic weights stored in low-bitwidth on-chip SRAM rather than requiring external DRAM (BrainChip Inc., 2023; BrainChip Inc., 2025). The architectural consequence that matters most for this project is that an NPU which receives no spike performs no dynamic computation and draws negligible power; on a sparsely-firing spiking network – which, as Section 4 shows, an EEG-derived signal typically produces, given how much of the recording is quiet interictal background – the majority of the chip’s NPUs are idle at any given moment, in direct contrast to a GPU, where every core is occupied for every tensor operation regardless of the input’s information content.

It is worth connecting this directly back to the physiology described in Section 2.1 above, since the two are not independent facts sitting side by side: interictal background – consistent with the 1–3% seizure-window class imbalance already noted in Chapter 3 – corresponds to the low-amplitude, loosely-correlated firing pattern described earlier, which produces comparatively sparse spiking activity on the chip and correspondingly low dynamic power draw; an ictal event corresponds to the highly synchronised, high-amplitude firing pattern that defines a seizure, which produces a denser burst of spiking activity and a genuine rise in dynamic power. This is not a coincidental match: it is the reason a sparse, event-driven architecture is a plausible fit for this specific problem,

rather than an arbitrary hardware choice applied to an unrelated task. The chip spends the overwhelming majority of a patient’s wear time idle precisely because that is also the overwhelming majority of a patient’s time spent seizure-free.

This power efficiency is not free. The AKD1000 v1 imposes a specific, hard set of architectural constraints that shaped every design decision in this project’s model architecture (Chapter 3), summarised in full in Table 2.1. Every model architecture variant considered in this project was designed to satisfy these constraints from the outset, rather than being adapted to them after an incompatible design had already been trained; the full constraints reference document is maintained alongside the project’s codebase at <https://github.com/samidR25/dissertation> for any constraint not covered by the summary below.

Table 2.1: AKD1000 v1 architectural constraints relevant to this project’s model design.

Constraint	Value	Consequence
Weight/activation bitwidth	Maximum 4 bits (w4a4)	Confirmed via <code>cnn2snn</code> ’s own model-compatibility validation as this hardware revision’s hard ceiling
Dilated convolutions	Unsupported (<code>dilation_rate</code> $\neq$ 1)	Fails at <code>check_model_compatibility()</code> ; excluded from every architecture, including longer-receptive-field designs considered early in the project
Batch normalisation	Must be folded or removed prior to conversion	No direct spiking equivalent exists; addressed via the BN-placement fix discussed in Chapter 3
Kernel sizes	Restricted to 1×1 , 3×3 , 5×5 , 7×7	Fixed silicon routing; larger or irregular kernels are not representable
Activation function	ReLU only	The ANN-to-SNN conversion (Section 2.4) is defined for ReLU and does not extend to LeakyReLU, GELU etc.
Skip connections	Not supported in early design iterations	Residual paths complicate conversion; simplified out of every architecture used for reported results
First-layer padding	<code>padding='valid'</code> required on first Conv2D	Workaround for a bug in <code>quantizeml</code> v1.2.3, not a modelling preference
Consecutive pooling	No two <code>MaxPool</code> layers may appear back-to-back	v1 hardware-mapping restriction; every pooling layer in <code>seizure_cnn_v2</code> is separated by at least one convolutional layer

2.4 ANN-to-SNN Conversion

The AKD1000 does not train spiking neural networks (SNNs) natively from scratch. Instead, the standard and the project’s own workflow is to train a conventional artificial neural network (ANN) with ReLU activations using ordinary backpropagation, then convert it into an equivalent spiking form for deployment. Maass (1997) first articulated SNNs as a distinct, biologically grounded computational paradigm – a “third generation” of neural network model, in which information is represented in the timing and rate of discrete spikes rather than continuous real-valued activations. Gerstner and Kistler (2002) give the standard mathematical treatment of the underlying single-neuron model most commonly used in this conversion, the Leaky Integrate-and-Fire (LIF) neuron, which accumulates incoming current on a membrane potential that leaks toward rest over time and emits a spike when that potential crosses a threshold, resetting immediately afterward. Formally, the membrane potential $V(t)$ evolves according to

$$\tau \frac{dV(t)}{dt} = -V(t) + R I(t) \quad (2.1)$$

where τ is the membrane time constant governing how quickly the potential leaks back toward rest, R is the neuron’s input resistance, and $I(t)$ is the incoming current – in a network context, the weighted sum of incoming spikes. Whenever $V(t)$ reaches a threshold V_{th} , the neuron emits a spike and $V(t)$ is immediately reset to a reset potential V_{reset} , after which the integration in Equation 2.1 begins again from that reset value. The practical consequence for this project is rate coding: a stronger, more sustained input current causes $V(t)$ to reach threshold more frequently, producing a higher spike rate, and it is this spike rate – not any single spike’s individual timing – that is taken to correspond to a ReLU unit’s continuous activation value in the conversion described below.

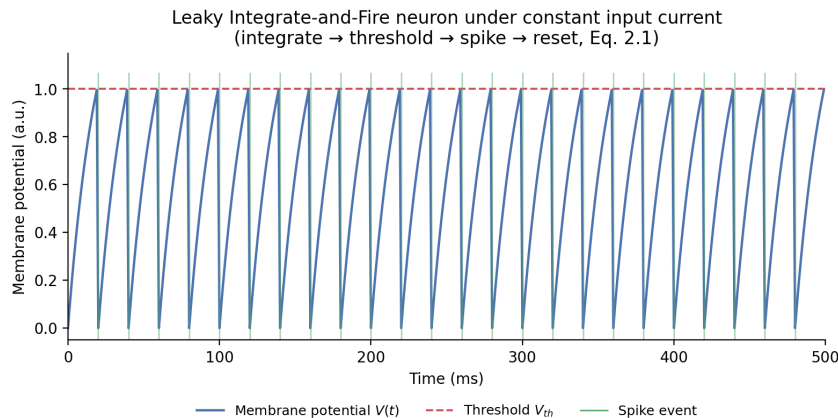


Figure 2.4: A simulated LIF neuron’s membrane potential over time under constant input current, showing the characteristic integrate-toward-threshold, spike, and reset pattern described by Equation 2.1.

The specific technique of converting an already-trained ANN into an SNN, rather than

training an SNN directly, is developed in detail by Rueckauer et al. (2017), and is the theoretical basis of the `cnn2snn` toolchain used throughout this project. The conceptual mapping is that a ReLU activation’s continuous output value is approximated by a LIF neuron’s spike *rate* over a short window of time – a higher ReLU activation corresponds to more frequent spiking, exactly the rate-coding relationship set out in Equation 2.1 above.

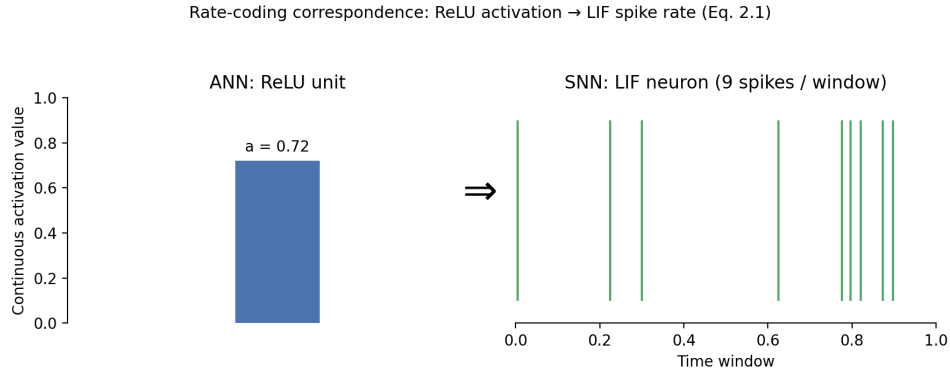


Figure 2.5: A conceptual before/after diagram: a ReLU unit’s continuous activation value on one side, mapped to a LIF neuron’s spike train over a short time window on the other, illustrating the rate-coding relationship the conversion relies on.

Pfeiffer and Pfeil (2018) give an accessible bridge between the classical LIF-neuron literature and this specific deep-learning conversion use case, and note explicitly that some accuracy loss through the conversion process (typically small, single-digit percentage points) is normal and expected, a trade-off this dissertation discusses candidly in Chapter 5 rather than treating any accuracy gap as evidence of an implementation error.

2.5 Related Detection Systems

2.5.1 Shoeb (2009)

Shoeb (2009) is the fair, established comparator for this project’s per-patient personalisation work (Section 4.3): a per-patient support vector machine classifier trained on each patient’s own labelled seizures, reporting a mean detection latency of approximately 4.6 seconds across the patients evaluated. This is a per-patient, not cross-patient, model – the correct comparator for a project section that is itself about the effect of a patient’s own data, not for the cross-patient LOPO work in Section 4.1.

2.5.2 Ali et al. (2024)

Ali, Angelova, and Karmakar (2024) train a random-forest classifier on 92 hand-crafted features per channel across 22 channels (748 features selected in total), reporting 72.63% sensitivity under subject-wise 5-fold cross-validation and 75.34% under a subject-wise

leave-one-out (LOO) protocol. Their Section 2.5 confirms the LOO protocol is a genuine cross-subject evaluation – “each of the 24 subjects was considered the test set, and the model was trained on the rest of the 23 subjects... the model was not tested with the data of any subjects it was trained on” – making it a fair, true zero-shot comparator for this dissertation’s own LOPO sweep (Section 4.1).

Ali et al. (their Section 2.7) do not implement anything equivalent to this dissertation’s collapse diagnostic. Their event-matching criteria are a $\geq 70\%$ (75% in their text) event-duration coverage threshold, a 10-second gap-merging rule, and discarding isolated detections under 10 seconds – no window-specificity floor, and no ceiling on how much of a recording a single contiguous predicted block may cover. Their own Table 2 makes the consequence of this concrete: several individual subject runs carry very high false-detection rates alongside reported perfect sensitivity – subject 5 at 15.03 FDR/hr, subject 6 at 14.15 FDR/hr, subject 8 at 10.80 FDR/hr – exactly the over-triggering pattern this dissertation’s collapse gate exists to catch and exclude from a headline sensitivity figure. Their reported 72–75% therefore very likely includes contributions from folds that would not pass this dissertation’s own collapse diagnostic; this is not a claim I am inferring indirectly any longer, but one their own published per-subject table supports directly.

A further comparability note, easy to miss without reading their methods closely: Ali et al. report a “false detection rate” (FDR), defined as a count of missed/false *events* per hour, which is not guaranteed to be an identical computation to this dissertation’s own FP/hr, even though both are hourly event-based rates – the two should be treated as similarly-named but not necessarily identically-defined metrics. Beyond the collapse-diagnostic difference, several other factors mean the two systems are not solving an identical problem under identical constraints: Ali et al.’s hand-crafted-feature random-forest approach assumes no hardware power, latency, or w4a4 quantisation constraint; and their 22-channel montage is wider than this dissertation’s unified 18-channel common montage (Chapter 3). The gap reported in Chapter 4 should be read with all of this in mind, not only the headline percentage difference.

2.5.3 Low-Power Embedded Seizure Detection

A separate family of prior work targets low-power embedded or wearable seizure detection directly, and one comparator is close enough to this project’s own setup to warrant a direct, cited comparison rather than a vague gesture at “similar” systems. Li et al. (2024) deploy a spiking neural network (the WaveSense architecture) on SynSense’s Xylo neuromorphic processor for seizure detection on the CHB-MIT dataset – the same dataset used throughout this dissertation – reporting a measured average power draw of $87.4\mu\text{W}$ (I/O) plus $287.9\mu\text{W}$ (compute core), for a total of approximately $375\mu\text{W}$, alongside 93.3% accuracy, 90.4% sensitivity, and 96.7% specificity in simulation (89.87% accuracy after on-

chip quantisation), and a median detection latency of 0.5 seconds.

Stated plainly, since this dissertation does not shy away from unflattering comparisons elsewhere: this reported figure is genuinely lower than this project’s own chip-level power estimate (approximately 1mW, Section 4.5). Some of that gap is explained by scope rather than by superior efficiency: Li et al. (2024)’s network uses 2.4 thousand weights against a narrower input (two electrode channels, C3-P3 and C4-P4, chosen specifically for their pronounced seizure signature, sigma-delta spike-encoded), whereas this project’s architecture (137,826 parameters, Chapter 3) processes the full 18-channel montage as a raw waveform input via a genuine spatio-temporal CNN rather than a narrower, pre-selected two-channel encoding.

Comparability is further limited by evaluation protocol: Li et al.’s reported accuracy/sensitivity/specificity figures are window- or segment-level, not the event-level sensitivity and FP/hr this dissertation treats as primary (Section 3.4), and their evaluation is per-patient rather than a cross-patient LOPO protocol. Although the primary difference is worth stating explicitly, since it bears directly on a principle this dissertation treats as foundational rather than optional: Li et al. partition their 5-second trials into training and test sets via a random 4:1 split, not a chronological one. This is precisely the leakage risk chronological splitting (Section 3) is designed to prevent – a random split can place one window from the middle of a seizure in training and a temporally adjacent, near-identical window from that same seizure in test, allowing a model to score well by recognising a near-duplicate rather than by genuinely generalising. This is not stated to dismiss their result, which remains a real, measured hardware deployment worth taking seriously, particularly on the power figure; it is stated because a reported accuracy sitting on top of a randomly-split evaluation is not directly comparable to a figure obtained under the chronological protocol used throughout this dissertation, and that difference should be visible to the reader rather than left for them to discover independently. Both projects’ hardware results are real, measured figures rather than datasheet estimates, which is the basis on which they are compared here at all – it reflects the differences in input scope, model capacity, and evaluation protocol, stated plainly rather than glossed over.

Other systems in this same design space – BrainFuseNet, EpiDeNet, and dedicated ultra-low-power accelerators more broadly – are not given the same direct treatment here, since a comparably careful, cited power/accuracy comparison was not completed for each within the scope of this dissertation; where referenced elsewhere in Chapter 5, they should be read as illustrative context for the broader energy-versus-accuracy design space this project occupies, not as benchmarked comparators in the sense Li et al. (2024) is here.

Chapter 3

Methodology

3.1 Dataset and Preprocessing

For this project I use the CHB-MIT paediatric scalp EEG dataset (Goldberger et al., 2000), restricted to the 18 channels common across all included patients’ recordings. Data is split chronologically, in a 70:15:15 ratio – the earliest 70% of each patient’s recording assigned to training, the next 15% to validation, and the final, latest 15% held out as test – rather than by conventional random stratification. The split configuration used throughout this project is fixed as a single, shared setting rather than re-derived per script:

```
split_ratios: tuple = (0.70, 0.15, 0.15)    # chronological train/val/test
```

Every result reported in Chapter 4 is evaluated against this split’s reserved, final 15% – saved and loaded throughout this project’s codebase as `X_test/y_test` – confirmed directly in `eval_event_level.py`, which refuses to run without an `X_test` array present in the dataset file it is pointed at.

This choice was deliberate and made because consider what a stratified (randomly shuffled) split would permit: a five-second window drawn from the middle of a given seizure event could land in the training set, while a temporally adjacent window from that same seizure event – sharing near-identical waveform morphology, since seizure activity smoothly builds up and is not disjointed in nature – could land in the test set purely by chance. A model evaluated this way could score well by effectively recognising a near-duplicate of something it had already seen, rather than by genuinely generalising to unseen seizure activity, and no real deployed device could ever benefit from this, since a device monitoring a patient in real time only ever has access to that patient’s past, never their future. Chronological splitting removes this possibility structurally: nothing in the test set can be a near-neighbour of anything in training, because everything in training necessarily precedes everything in test in time. I rejected stratified splitting for

this reason, and I consider chronological splitting a required baseline for methodological honesty in this problem, not merely one reasonable option among several.

3.1.1 Signal Preprocessing

Each recording is preprocessed independently, file by file, before any windowing or splitting: a 0.5–40Hz bandpass filter (FIR, Hamming window) removes slow DC drift and high-frequency noise outside the range relevant to ictal activity; a 60Hz notch filter removes US mains interference (CHB-MIT was recorded at Boston Children’s Hospital, hence 60Hz rather than the 50Hz appropriate for UK recordings and the evidence motivating this filter is given in Figure 3.1); and a Common Average Reference (CAR) – subtracting the mean of all channels from each individual channel – reduces noise common to every electrode, such as movement artefact or shared electrical interference. Recordings are resampled to 256Hz where necessary, though this is a no-op for CHB-MIT, which is already recorded at that rate; the resampling step is kept explicit in the pipeline regardless, for reproducibility if this pipeline is ever applied to a differently-sampled dataset.

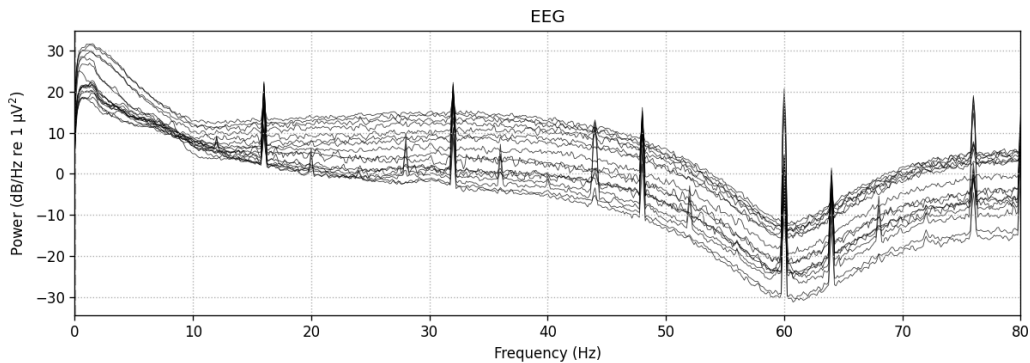


Figure 3.1: Power spectral density across all 18 channels of a sample CHB-MIT recording, computed during exploratory data analysis. The sharp peaks at 60Hz and its harmonics are consistent with US mains electrical interference (CHB-MIT was recorded at Boston Children’s Hospital) – this observation directly motivated the 60Hz notch filter described above.

Two further implementation details are worth stating explicitly, since both reflect real bugs avoided rather than arbitrary choices. First, channels are selected by *index* (positions 0–17), not by name: chb01’s recording contains two channels both literally named T8-P8 (MNE renames the second on load to avoid a naming collision), and a name-based channel lookup would silently select the wrong one or crash depending on load order. Verified against all 22 patients’ summary files, positions 0–17 correspond to the same 18 physical channels across every patient, making index-based selection both correct and consistent. Second, EDF files are never concatenated across files before windowing: CHB-MIT recordings for a given patient are split across multiple files with a real hardware gap of approximately 10 seconds between them, and treating this as if it were continuous

signal would create a false continuity across the gap, corrupting any window that happened to straddle a file boundary. Every window in this dissertation’s results is built from a single EDF file’s data only.

Each recording is then divided into fixed 2-second windows with a 1-second stride (50% overlap) to maximise training density. To ensure label integrity, windows are designated positive only when $\geq 50\%$ of their duration overlaps an annotated seizure. This criterion removes transitional onset/offset windows with ambiguous ground truth, to establish a clean, unambiguous decision boundary.

10-20 electrode positions used to form the 18 common bipolar channels
(19 electrodes in a chain montage $\rightarrow$ 18 pairs, e.g. FP1-F7, F7-T7; crossed out = FT9/FT10, the only genuinely extra electrodes verified against a real chb01_01.edf dump -- the other 3 excluded channels re-pair existing electrodes: P7-T7, FT10-T8, T8-P8[dup])

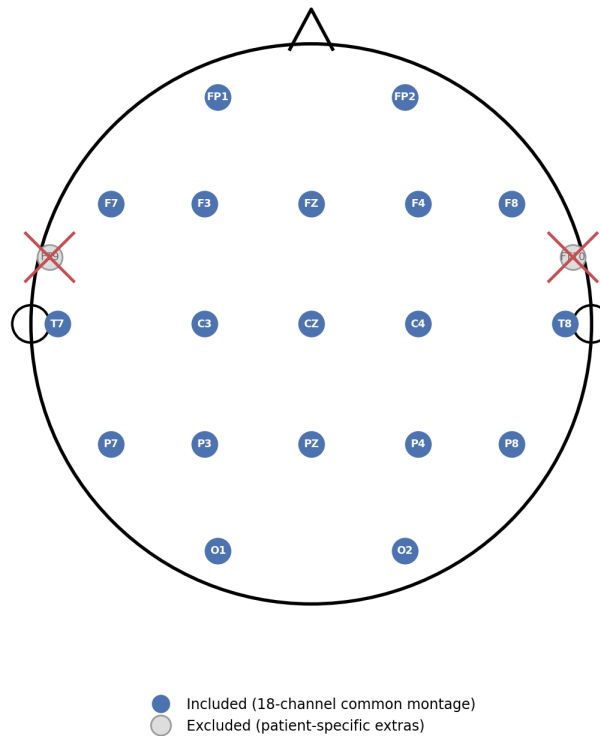


Figure 3.2: Standard EEG electrode placement (10-20 system) for the 18-channel montage used in this project. Verified against a real channel dump from `chb01_01.edf`: only FT9 and FT10 are genuinely extra electrodes outside the 18-channel set; the remaining three nominally "excluded" channel labels (P7-T7, FT10-T8, and a duplicate T8-P8) are re-pairings of electrodes already present in the montage, not additional physical electrode sites.

3.1.2 Chronological Splitting and SMOTE Ordering

Given the low prevalence of seizure-positive windows relative to background EEG (typically 1–3% of windows for a given patient), I apply class-imbalance correction via SMOTE (Synthetic Minority Over-sampling Technique) to the training portion of each split. It is worth explaining candidly how I arrived at the specific ordering used throughout this

dissertation, because it was not obvious to me from the outset, and I do not want to present the final, correct version as though it had been the plan from day one.

Early in this project, I applied SMOTE to each patient’s full dataset *before* carving out the chronological split – generating synthetic minority-class (seizure) samples across the whole recording, then splitting the resulting, already-augmented dataset into train/validation/test. It was only once I was auditing intermediate outputs for an unrelated check that I noticed synthetic samples generated from what should have been strictly training-only data were structurally capable of having near-identical neighbours land across the split boundary in validation – meaning a portion of what I was calling “unseen validation performance” could, in principle, reflect the model recognising a synthetic near-duplicate of something it had implicitly been shown, rather than genuine generalisation. I corrected this project-wide once I found it. SMOTE is now applied *only* to the already-separated post-split training portion throughout every result reported in this dissertation, never to validation or test data, and never before the chronological split itself is made.

The undersampling ratio applied before SMOTE differs between single-patient and pooled builds, and this difference is deliberate rather than an inconsistency: single-patient builds (like C2) undersample the majority (non-seizure) class to a **10:1** ratio relative to the minority class before SMOTE rebalances it further, while pooled/multi-patient builds (pool7, pool6-minus-X, and every LOPO fold) use a tighter **5:1** ratio, applied *per domain* (i.e. per contributing patient) before pooling. The tighter 5:1 ratio for pooled builds was a direct response to the compute constraints described in the next subsection: a 10:1 ratio applied per-patient across seven or more pooled patients produced a combined training array too large to reliably fit in memory during SMOTE’s own internal neighbour-search step, and 5:1 was the ratio found to keep pooled builds within the available RAM without materially changing the class-imbalance-correction principle at work.

3.1.3 Development Environment Constraints

It is worth stating plainly, since it is not otherwise visible anywhere in this dissertation’s results, that every design decision above – and several architectural and experimental-scope decisions elsewhere in this chapter – were made under binding compute constraints, not chosen freely from an unconstrained menu of options. All development, training, and dataset construction for this project ran on a single laptop via WSL2 (Windows Subsystem for Linux), with approximately 11GB of RAM available to the environment and a shared disk budget that had to be actively monitored during larger multi-patient builds.

This constraint shaped concrete decisions documented elsewhere in this dissertation, not just as background colour: the SMOTE undersampling ratios above were tuned specif-

ically to fit within this RAM budget; the pool7/pool6-minus-X experiments (Section 4.2) were run as a bounded, seven-patient expansion rather than a larger pool, in part because dataset-building time and memory footprint scale with pool size on this hardware; and an out-of-memory bug in the pool-dataset-building script (an unused array construction consuming approximately 6.85GB unnecessarily) reliably crashed builds on this machine before it was identified and fixed, a bug that would very plausibly not have surfaced at all on a machine with a larger RAM budget. Training run duration was also a live constraint on experimental scope: individual LOPO folds and multi-patient pool builds took long enough (multiple hours per full sweep) that iteration speed was a genuine factor in which candidate techniques were screened at a single seed/patient before committing to a fuller multi-seed or multi-patient confirmation run, a screening discipline described further in Section 4.3 and Chapter 5.

3.2 Model Architecture

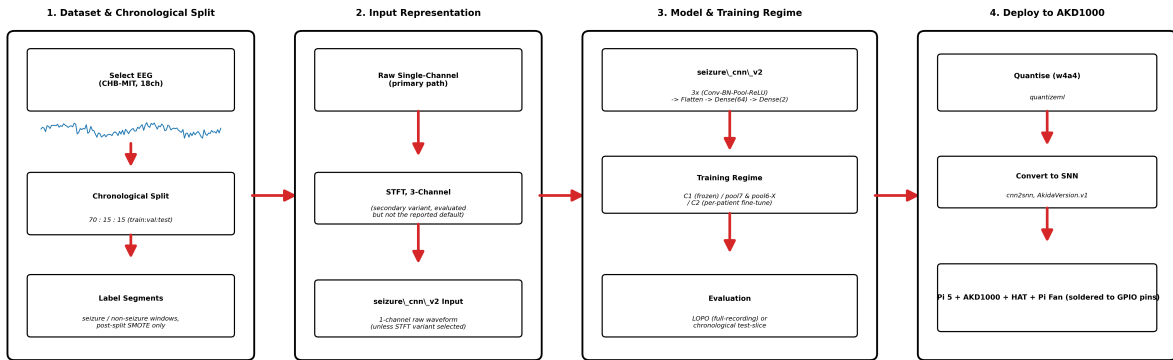


Figure 3.3: An end-to-end pipeline diagram: raw scalp EEG input, preprocessing and chronological splitting, the seizure_cnn_v2 architecture (with layer-by-layer block structure), w4a4 quantisation, ANN-to-SNN conversion via cnn2snn, and final deployment onto AKD1000 v1 silicon, annotated at each stage with the relevant section number in this chapter.

3.2.1 Spatio-Temporal CNN (`seizure_cnn_v2`)

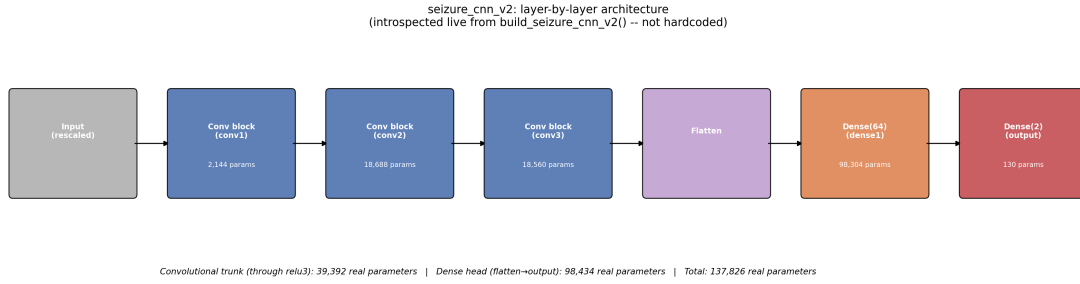


Figure 3.4: Layer-by-layer diagram of `seizure_cnn_v2`: three repeated Conv-BN-Pool-ReLU blocks, followed by Flatten, Dense(64), and Dense(2, softmax). Generated dynamically from the real `build_seizure_cnn_v2` definition, reading actual per-layer parameter counts rather than hardcoded figures.

The base architecture, referred to throughout as `seizure_cnn_v2`, is a compact spatio-temporal convolutional neural network. It is worth describing its three convolutional blocks precisely rather than as three identical repeated units, since they are not: the first block uses a deliberately large, asymmetric kernel – 32 filters at (9,7) with stride (1,4) – chosen to capture broad spatio-temporal structure across both the channel and time axes in a single layer, followed by (1,2) max pooling; the second block uses a more conventional 64-filter (3,3) kernel with (2,2) pooling; the third block reduces back down to 32 filters at (3,3) with (2,2) pooling, giving the network a $32 \rightarrow 64 \rightarrow 32$ bottleneck filter-count shape rather than a monotonically increasing one. Each convolutional block is followed by batch normalisation, then the pooling and ReLU activation described above. The network concludes with a flatten operation and two dense layers (64 units, no bias, then 2 units with softmax output for the binary seizure/non-seizure decision).

Two further architectural details reflect some toolchain constraints rather than design preferences: the first convolutional layer uses `padding='valid'` specifically as a workaround for a bug encountered in `quantizeml` version 1.2.3, and no two max-pooling layers are ever placed consecutively, a constraint imposed by the AKD1000 v1 hardware mapping rather than a modelling choice.

Parameter counts are confirmed directly against the model’s own `model.summary()` output: contains 137,826 total parameters (39,392 in the convolutional trunk; 98,434 in the dense head). While Keras reports a unquantised `float32` footprint of 538.38 KiB, the compiled `w4a4` spiking model binaries (`.fbz`) consistently measure 166,450 bytes (≈ 162.5 KiB) on disk across all training regimes. This footprint exceeds theoretical 4-bit weight storage (≈ 68.9 KiB) due to container format overhead, layer metadata, and full-precision bias storage. I implemented and evaluated both a single-channel raw-EEG input variant and a three-channel short-time Fourier transform (STFT) variant; the raw single-

channel variant is the one used for the results reported in Chapter 4 unless otherwise stated.

Placing Batch Normalisation directly after each convolutional layer was critical as alternative orderings caused representational collapse during ANN-to-SNN conversion. The final design incorporates this fix.

3.2.2 Training Procedure

All training in this project – C1, pool7/pool6-minus-X, and the base checkpoint from which C2 fine-tuning starts – uses a consistent procedure: the Adam optimiser, sparse categorical cross-entropy loss, a batch size of 32, early stopping on validation loss (patience 15 epochs), and a learning-rate reduction schedule (halving on plateau, patience 7 epochs, down to a floor of 1×10^{-7}). Class imbalance is corrected via SMOTE at the data level (Section 3.4 above) *and*, separately, via explicit class weighting during training (the seizure class weighted $1.5\times$ relative to the non-seizure class) – these two corrections are applied together, not as alternatives to one another as one does not supersede the other.

Reproducibility is handled by explicit, deterministic seeding at the start of every training run (`random.seed`, `np.random.seed`, and `tf.random.set_seed`, all set from the same value; default 42, though the canonical C2 results reported in Chapter 4 use seed 256 specifically). The seed-sensitivity discussion in Section 4.3 concerns genuine variance in outcome *across* different seed values, not a lack of within-run determinism as each individual run is fully reproducible given its seed.

Additionally, the early-stopping validation subset used during training is not simply the chronological validation portion from the 70:15:15 split, because doing so carries an identified risk for at least one patient, addressed proactively rather than only after being observed to go wrong. chb03’s chronological validation portion happened to contain zero seizure windows (all three of chb03’s annotated seizures fell entirely within the chronological training or test portions), which meant early stopping monitored against that chronological validation split for this patient would have been driven by just noise rather than any genuine seizure-relevant signal, had the naive approach been used. In contrast to this chb10, never posed this risk, because its chronological validation portion happened to contain 66 seizure windows. The fix applied throughout this project stratifies a seizure-guaranteed validation subset *from within the already-isolated training portion only*, used exclusively for early-stopping and checkpoint selection during training.

This is stated explicitly because it must not be confused with violating the chronological-splitting principle, described at the start of this chapter: every result reported in Chapter 4 is still evaluated exclusively against the untouched, fully chronological `X_test` split, confirmed directly in `eval_event_level.py`, which refuses to run without it:

```
assert 'X_test' in data.files, f"{data_path} has no X_test split"
```

```
X_eval = data['X_test'].astype('float32', copy=False)
```

and the stratified carve-out itself, in `train_baseline.py`, is explicit in its own docstring that it operates only on the already-isolated training pool, never on validation or test data:

```
def _stratified_real_split(data, seed=42, test_size=0.20, ...):
    """
    Splits the REAL, undersampled, pre-SMOTE train pool (X_train_real/
    y_train_real) stratified, so val is GUARANTEED to contain seizure
    windows regardless of how they fall chronologically. SMOTE applied
    only to the resulting train portion. Val is real, untouched -- used
    ONLY for early-stopping/checkpoint selection, never reported as a
    held-out metric (that's still the chronological X_test).
    """
```

The stratified carve-out affects only which checkpoint gets selected during training, never what that checkpoint is subsequently evaluated against.

3.2.3 Quantisation (w4a4) and Hardware Constraints

Prior to conversion, the trained ANN is quantised using BrainChip’s `quantizeml` toolchain to four-bit weights and four-bit activations (w4a4), with per-tensor activation quantisation parameters mandatory for compatibility with the target hardware. This four-bit ceiling is not a design preference adopted for convenience; it is confirmed, via the `cnn2snn` toolchain’s own model-compatibility validation, to be the hard architectural ceiling of the AKD1000 v1 revision targeted by this project (Table 2.1, Chapter 2). Similarly, dilated convolutions (any layer with `dilation_rate` $\neq 1$) are unsupported by this hardware revision and were excluded from every architecture variant I considered, including longer-receptive-field variants explored early in the project and since abandoned for this reason.

3.2.4 ANN-to-SNN Conversion

Conversion from the trained, quantised ANN to its spiking equivalent is performed via BrainChip’s `cnn2snn` toolchain, following the conceptual approach set out by Rueckauer et al. (2017) and the LIF formulation given in Equation 2.1 (Chapter 2). Every conversion call in this project’s codebase is made within an explicit `AkidaVersion.v1` context manager:

```
with set_akida_version(AkidaVersion.v1):
    akida_model = cnn2snn.convert(quantised_model)
```

ensuring the conversion targets the correct hardware generation rather than silently defaulting to an incompatible one. This is a standing project rule, not a response to a specific documented incident: `cnm2snn`’s `convert()` defaults to targeting Akida 2.0 rather than the AKD1000’s v1 hardware if this context manager is omitted, and doing so would fail silently rather than raising an obvious error.

3.3 Training Regimes

3.3.1 C1 — Frozen Shared Pool (chb01/02/05)

The foundational training regime, referred to throughout as C1, trains a single shared model once on a fixed three-patient pool (chb01, chb02, chb05) and evaluates it cold against other patients, with no further adaptation. It is important to state C1’s actual role in this project precisely, since it would be easy to over-read its purpose due to its small size and simplicity: C1 was not built or used as this dissertation’s primary comparative result. Its function was largely infrastructural – establishing a working, reproducible baseline used as a fast, low-cost testbed throughout this project’s development (several of the methodological errors discussed in Chapter 5 were caught during earlier stages of this project, though it was not every case that C1-based runs specifically were each was first noticed, rather than other parts of the pipeline), and providing the starting checkpoint from which the pool was later scaled up in stages (to four, then seven, then eight patients, as described below and in Chapter 4). Where C1 does appear in later comparative tables (Section 4.3), it is specifically as the smaller-pool reference point in a *pool-size* comparison, not as this dissertation’s central performance claim.

C1’s deliberately small scope reflects a working principle I had aimed to follow throughout this project, given the compute constraints described in Section 3.4 above, every new technique or pipeline change was first screened on the smallest, cheapest configuration that could plausibly reveal whether it worked at all – a single seed, a small patient set, a short training run – before committing the time and memory budget to a fuller multi-seed or multi-patient confirmation. C1’s three-patient scope was the first instance of this “fail cheaply, iterate quickly” discipline. A three-patient training run is fast enough to iterate on repeatedly, which is the practical value of starting there rather than at a larger pool size, regardless of which specific issues were or weren’t caught at that scale in practice. The same ethos is acted upon again later in this project’s development, in the single-seed, small-patient pilots used to screen the domain-adaptation candidates in Section 4.4 before any of them were ruled out or confirmed at fuller scale.

3.3.2 Pool7 and pool6-minus-X — The Primary Own-Data-Inclusion Probe

This dissertation’s primary evidence for the effect of personalisation is drawn from a different training regime with a seven-patient frozen pool (chb01, chb02, chb05, chb06, chb07, chb09, chb20; referred to as pool7), evaluated against each of its own constituent patients’ full recordings. This was done to isolate whether a patient’s *own data being present in the pool* is what drives any observed benefit, as opposed to the pool simply being larger. In turn, I also trained a matched six-patient pool excluding each target patient in turn (pool6-minus-X) and evaluated it zero-shot against that same excluded patient, holding pool size approximately constant across the comparison (six patients versus seven) and varying only the one factor of interest. This pool6-minus-X-versus-pool7 comparison, detailed in Section 4.2, is now this dissertation’s leading evidence for the general claim that a patient’s own data measurably improves their own detection outcome. It is worth stating plainly that this is a related but mechanistically distinct claim from the C2 fine-tuning regime described next: pool7/pool6-minus-X demonstrate this effect via *pool composition* on an otherwise still-frozen shared model, whereas C2 demonstrates a related effect via explicit *per-patient fine-tuning*. The two should not be read as the same experiment conducted at different scales.

3.3.3 C2 — Per-Patient Fine-Tuning

A second, narrower personalisation-related regime, referred to as C2, adapts the C1 model to each of a canonical set of four personalisation-eligible patients (chb10, chb13, chb15, chb16) individually via a **three-phase gradual unfreeze**, not a single-shot freeze-trunk-train-head procedure (bolded to not misstate which layers are actually trainable at each stage). Phase 1 trains only the dense head and the batch normalisation layers (every convolutional layer frozen), for 20 epochs at a learning rate of 1×10^{-4} . Phase 2 additionally unfreezes the third convolutional block, for a further 20 epochs at 5×10^{-5} . Phase 3 additionally unfreezes the second convolutional block, for 30 epochs at 1×10^{-5} . The first convolutional block – the large (9, 7) spatio-temporal kernel described above – remains frozen throughout all three phases, on the reasoning that this layer’s broad structural features are expected to transfer across patients even where later, more specific layers do not. Learning rate decreases and epoch budget increases across the three phases by design, reflecting progressively more of the network becoming trainable at each stage. Batch normalisation layers being trainable even in Phase 1, despite every convolutional layer being frozen at that stage, was itself a specific fix applied during this project’s development, not the original design.

C2-related results (Section 4.3) primarily support a narrower discussion of pool-size and training-seed effects, rather than serving as the central personalisation evidence, a

role held by Section 4.2. The fine-tuning result itself, and the conformal-prediction-based correction to its reported false-positive rate, remain genuine, fully reported findings.

3.4 Evaluation Protocol

3.4.1 Full-Recording LOPO vs. Chronological Test-Slice Evaluation

Two evaluation regimes are used in this project, and distinguishing them precisely is essential, not a formality: conflating them was the root cause of a real, caught-and-corrected error in this project’s own history (below). The first, full-recording leave-one-patient-out (LOPO) evaluation, trains on all patients except the one being evaluated and tests on that patient’s *entire* recording, since that patient contributed zero data to training in that fold. This is the only evaluation regime in this dissertation that constitutes a genuine cross-patient generalisation test, and the only regime fairly comparable against Ali, Angelova, and Karmakar (2024)’s LOPO figures. The second regime, used elsewhere in the pipeline, for instance for the C2 patients, who *do* contribute training data via their own fine-tuning split, evaluates only against a smaller, held-out chronological segment of that same patient’s own recording, not their entire recording.

These two regimes answer different questions and are not interchangeable. My own LOPO cross-patient sweep was, at one stage, silently re-run using the smaller chronological-slice evaluation set for all fifteen patients rather than each patient’s full recording – an error I caught before any figure derived from that run was finalised, and corrected via an explicit full-recording evaluation flag. I report that correction explicitly, as per my standing practice of disclosing caught errors rather than omitting them.

3.4.2 Event-Level vs. Window-Level Metrics

As introduced in Chapter 2, event-level sensitivity and false-positive rate per hour (FP/hr) are treated as this dissertation’s primary reporting metrics, on the grounds that a clinician or patient cares about whether a genuine seizure *event* was flagged, not the count of individual, sub-second windows classified correctly within it. Window-level metrics, particularly window specificity, remain in use as a diagnostic tool. Stating “event-level” as the primary unit of reporting is not itself sufficient, however, without specifying precisely what constitutes one detected event from a raw sequence of per-window predictions is a choice with many potential alternatives. All of which is set out in full below.

What Constitutes a Detected Event

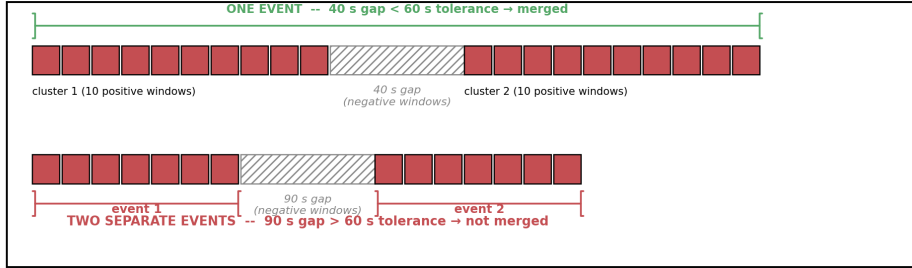
Two questions have to be answered before a sequence of window-level positive/negative predictions can be turned into a single event-level sensitivity and FP/hr figure: which nearby positive windows count as the *same* event rather than separate ones, and how much of an event’s duration needs to be correctly flagged before that event counts as detected at all.

Event grouping. Positive-predicted windows within 60 seconds of one another are merged into a single event, rather than counted separately. This value was arrived at empirically, not assumed: an earlier, tighter 10-second gap tolerance was tested on chb10 and found to fragment a run of false-positive windows into 636 separate spurious “events” (84.77 FP/hr) – clearly too aggressive a grouping rule. Widening the tolerance to 60 seconds collapsed this to 68 events (9.06 FP/hr) on the same data, with true event sensitivity completely unaffected by the change. This 60-second value is also independently grounded in clinical practice: the ILAE’s convention for seizure clustering treats ictal discharges that fall below threshold and recover within 60 seconds as part of the same seizure cluster, not as separate events, in commercial long-term EEG monitoring (Fisher et al., 2014).

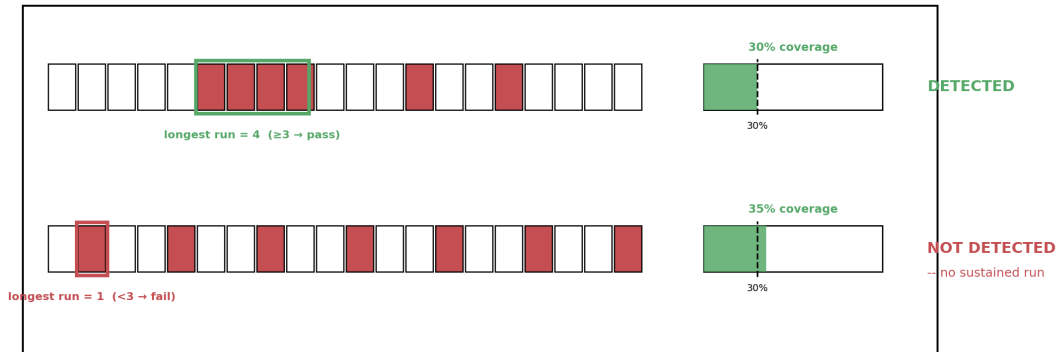
Detection criterion. An event counts as detected only if *both* of the following hold: at least 30% of its windows are predicted positive, and there is a minimum sustained run of 3 consecutive positive-predicted windows (approximately 3 seconds, at the 1-second window step used throughout this project). Neither threshold alone was judged sufficient. The 30% figure reflects a biological reality rather than an arbitrary convenience: seizures have a recruitment phase at onset (roughly the first 2–5 seconds) during which the EEG signal is often still sub-threshold for a window-level classifier trained predominantly on fully-developed ictal activity, so requiring near-total window coverage across an event would penalise the model for a phase of the seizure it was never realistically going to catch; catching the propagation phase once the seizure is fully underway is treated here as clinically sufficient. The 3-window minimum sustained-run requirement exists to prevent a handful of scattered, isolated positive windows – which could occur by chance even at a 30% overall rate – from counting as a genuine detection. This value was chosen to sit below CHB-MIT’s own shortest annotated seizure duration (7 seconds), so that genuinely short seizures remain detectable in principle. Furthermore, it sits comfortably below the conventional 10-second duration threshold widely adopted in clinical electrographic seizure definition and commercial long-term video-EEG monitoring systems (Tatum et al., 2022), balancing early event capture against transient artefact rejection.

Event-formation criteria: grouping and detection

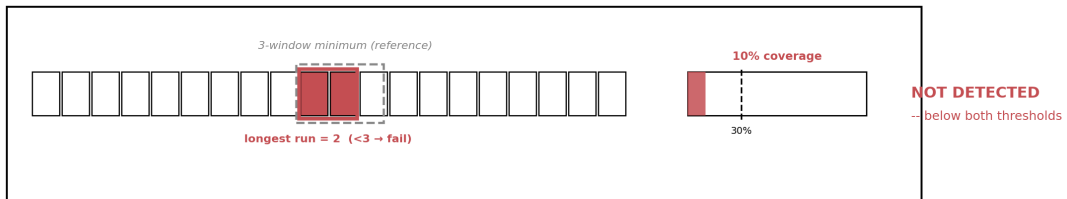
A. Event Grouping (60 s gap-tolerance rule)



B. Detection Criterion ($\geq 30\%$ window coverage AND ≥ 3 consecutive positive windows)



C. Near-Threshold Failure (short burst clears neither criterion)



Event grouping (60 s gap tolerance) and the joint detection criterion ($\geq 30\%$ window coverage AND ≥ 3 consecutive positive windows).

Figure 3.5: Event-formation criteria illustrated: (a) the 60-second gap-tolerance rule merging or separating nearby positive-window clusters into one or two events; (b) and (c) the joint detection criterion ($\geq 30\%$ window coverage *and* ≥ 3 consecutive positive windows), showing both a detected event and two ways an event can fail to qualify.

Detection latency, reported alongside sensitivity and FP/hr throughout Chapter 4, is measured directly from this same procedure: the time from an event's true onset to the

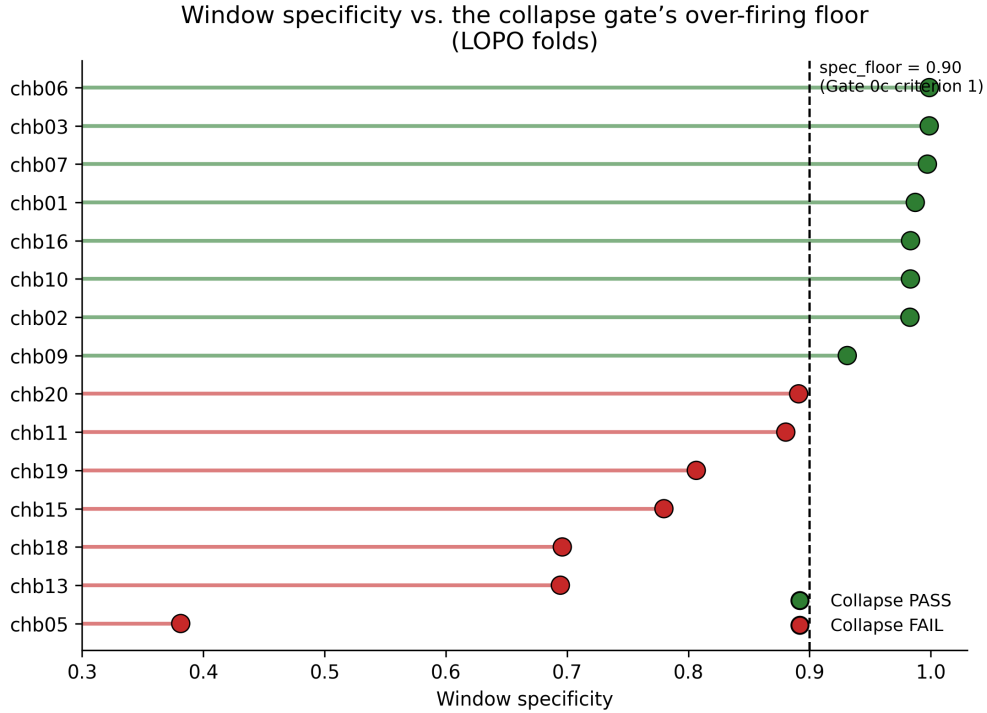
start of its first sustained qualifying run, not a separately computed figure.

3.4.3 The Collapse Diagnostic

A model that fires on almost every window it sees will register a high event sensitivity by simple virtue of never staying silent through a genuine seizure. This is an artefact of near-permanent triggering, not genuine discrimination. I built the collapse diagnostic specifically to catch this, and it checks three explicit conditions; a fold or patient result is only reported as a genuine "pass" if *none* of the following trigger:

1. **Window specificity below 0.9.** I set this floor because a model firing on more than 10% of genuinely non-seizure windows is, in my judgement, already behaving closer to an alarm that is near-permanently active than to a discriminating detector, regardless of how high its reported sensitivity looks.
2. **A single contiguous predicted block covering more than 90% of the recording.** This catches the specific over-triggering pattern that motivated this diagnostic in the first place: a model that fires once and simply never stops, which technically overlaps every genuine seizure event (hence a sensitivity of 1.0) without ever having discriminated anything.
3. **Positive-window fraction below 10% of the true seizure-window rate.** This is a mirror-image failure where the model that has effectively gone silent, firing so rarely that its few detections are more plausibly noise than genuine discrimination. Here I am simply enforcing an under-firing collapse that the first two conditions alone would not.

The canonical illustration of why this gate is necessary, referenced repeatedly throughout Chapter 4, is chb13 under the C1 frozen-pool model, whose apparently perfect event sensitivity was in fact almost entirely condition (1) and (2) above: a single, near-permanent predicted block covering nearly the whole recording. The three conditions above are implemented directly as follows:



Gate 0c FAILs a fold if ANY of: (1) window specificity < 0.90, (2) largest predicted block covers >90% of the recording, or (3) positive-window fraction < 10% of the true seizure rate (under-firing). In this LOPO run, all 7 FAIL folds happen to cross the specificity floor alone (criterion 1) — criteria 2/3 exist to catch failure modes criterion 1 would miss, but were not needed to explain any FAIL here.

Figure 3.6: Three panels illustrating the collapse diagnostic's three conditions: (a) a normal, discriminating prediction trace; (b) an over-firing collapse (a single near-permanent predicted block); (c) an under-firing collapse (almost no detections).

```
def collapse_diagnostic(y_pred, window_specificity, y_true=None,
                        step_s=1.0, gap_tolerance_s=60.0,
                        spec_floor=0.9, block_frac_ceiling=0.9,
                        underfire_ratio_floor=0.1):
    # Condition (1): window specificity below the floor
    if window_specificity < spec_floor:
        reasons.append(f"window specificity {window_specificity} < {spec_floor}")
    # Condition (2): a single block covers too much of the recording
    if largest_block_frac > block_frac_ceiling:
        reasons.append("largest predicted block too large")
    # Condition (3): under-firing relative to the true seizure rate
    if pos_frac < underfire_ratio_floor * true_frac:
        reasons.append("positive-window fraction too low -- likely under-firing")
    return {"pass": len(reasons) == 0, "reasons": reasons, ...}
```

3.4.4 Primary Metric: AUPRC over ROC-AUC

Given the 1–3% seizure-window class imbalance typical across CHB-MIT patients, I treat the area under the precision-recall curve (AUPRC) as this dissertation’s primary window-level discrimination metric, rather than the area under the ROC curve (ROC-AUC), which can present a misleadingly favourable picture on heavily imbalanced data. This is due to a classifier being right about the (rare) positive class only a small fraction of the time can still achieve a deceptively high ROC-AUC, since ROC-AUC’s false-positive-rate axis is computed against the (very large) negative class, diluting the effect of missed positives. AUPRC does not have this property as both its axes are computed with respect to the positive class.

I include it because it is the convention most widely used across the broader machine-learning and clinical-classification literature, including several of the comparator systems and design-space references discussed in Chapter 2, and reporting it allows this dissertation’s results to be cross-referenced against that literature directly. As such, it should be read as additive to AUPRC, not a replacement for it as this dissertation’s primary metric.

The justification for this ordering is not just the abstract imbalance argument; it rests on the back of a correction encountered during this project. A base-rate-relative AUPRC skill ratio – a patient’s raw AUPRC divided by their own seizure-window base rate, correcting for the fact that raw AUPRC is not directly comparable across patients with different prevalence – revealed that chb15’s C1 ratio was $6.0\times$ base rate (indicating genuinely decent discrimination, with the model’s threshold simply misplaced), while chb13’s ratio was only $1.3\times$ (near-chance, indicating genuinely weak discrimination). This is close to the *opposite* of the hypothesis I began this project with, which expected chb15 to be genuinely inert and chb13/chb16 to be cases of misplaced thresholding. Separately, across the full 15-fold LOPO sweep, some patients’ ROC-AUC values look superficially reasonable in isolation while the base-rate-corrected AUPRC reveals near-chance discrimination once prevalence is accounted for. Both observations are stated here as justification for this metric choice, rather than relying on the imbalance argument alone.

3.4.5 Conformal Prediction for FP/hr Calibration

Sweeping operating thresholds over the test set to report false-positive rates (FP/hr) introduces subtle data leakage, which can produce overly optimistic performance figures (discussed further in Section 4.3). To eliminate this post-hoc tuning pitfall, I apply *split-conformal prediction* to calibrate operating thresholds.

Calibration uses each patient’s chronological validation split which is data already set aside for early stopping, thus requiring no additional data collection or model re-training. Under the assumption of exchangeability within a single patient’s recording, split-conformal prediction yields an exact, distribution-free coverage guarantee:

$$P(\text{a future non-seizure window triggers the system}) \leq \alpha \quad (3.1)$$

Throughout this project’s conformal-calibrated evaluations, α is set to 0.01. This guarantees that no more than 1% of genuinely non-seizure windows will trigger a false alarm on unseen test data. Crucially, this calibration is applied strictly on a per-patient basis (like for C2) where intra-patient exchangeability holds; it is deliberately not extended cross-patient, where that assumption breaks down.

3.5 Hardware Deployment

Deployment used a Raspberry Pi 5 fitted with a BrainChip AKD1000 v1 M.2 card via the original manufacturer M.2 HAT+, connected over a direct static-IP link with no DHCP, and a 5V/0.12A cooling fan soldered directly to the Pi’s GPIO header to manage thermal load during sustained inference runs.

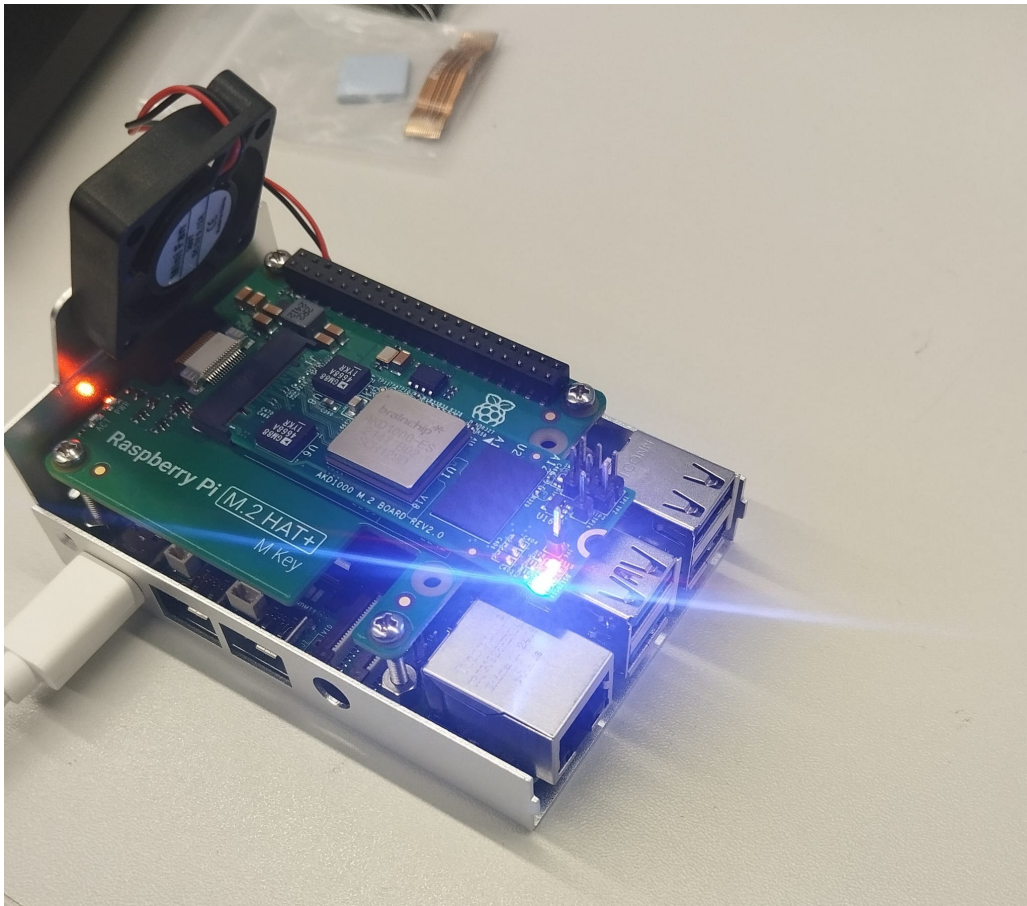


Figure 3.7: The working physical deployment: Raspberry Pi 5, AKD1000 v1 M.2 card and HAT+, and the GPIO-mounted cooling fan, as used for every hardware result reported in this dissertation.

A carrier-board substitution (a Pineboards HAT AI Bundle, attempted mid-project for

improved physical clearance) failed to enumerate the AKD1000 card at all. I isolated this to a fault in the replacement board or its flexible printed circuit cable, not the Raspberry Pi itself or the AKD1000 card – confirmed via `lspci` showing no device present on the replacement board where the original board correctly enumerated the AKD1000 – and reverted to the original, confirmed-working M.2 HAT+ for all hardware results reported in this dissertation.

3.5.1 Power Measurement Methodology

Two genuinely different power figures are reported in this dissertation, and neither is presented as validating or superseding the other. The first is a chip-isolated estimate (approximately 1mW active power), derived from BrainChip’s own datasheet figures for the AKD1000’s isolated neuromorphic compute, and used in this dissertation’s energy-per-inference calculations (Section 4.5).

The second is a system-level bench measurement, taken directly on the physical deployment described above.

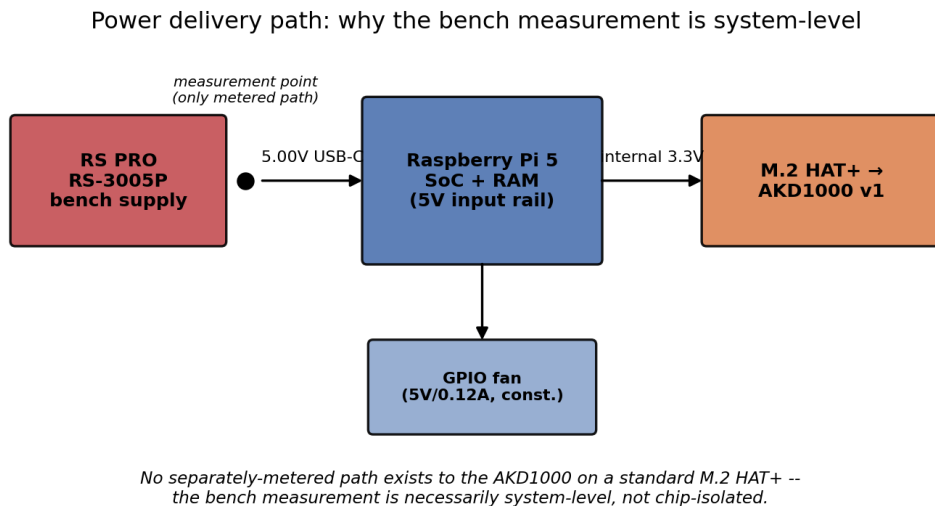


Figure 3.8: Block diagram of the Pi 5’s 5V USB-C input, the internally-derived 3.3V rail feeding the HAT+/AKD1000 card, and the single external measurement point, showing why the bench measurement is necessarily system-level rather than chip-isolated.

An RS PRO RS-3005P bench DC power supply was operated in constant-voltage mode (5.00V, 5A current limit), connected directly to the Raspberry Pi 5’s USB-C power input, with current read at an idle state (the trained model mapped to the device but no inference running) and an active state (a sustained inference loop running continuously for 30 seconds). This was repeated across three trials, full results for which are given alongside the corresponding measurement in Section 4.5.

This measurement is unavoidably a system-level figure, not a chip-isolated one. This is because the AKD1000 M.2 card's own 3.3V supply is derived internally from the Raspberry Pi's single 5V input rail, and no separately-metered power path to the card exists on a standard M.2 HAT+. The bench supply therefore necessarily measures the combined draw of the Raspberry Pi 5's own system-on-chip, its RAM, the HAT+, the AKD1000 card, and the continuously-running GPIO-mounted cooling fan shown in Figure 3.7, all together, at the single point where power enters the whole system. A genuinely chip-isolated measurement would require a shunt resistor inserted directly into the PCB trace feeding the M.2 connector, read with a precision instrument; I assessed this and deliberately did not attempt it, given the risk of physical damage to the hardware this close to the project's submission deadline, and given that the AKD1000 SDK exposes no run-time power telemetry on this hardware revision (`device.statistics` returns empty) that might have offered a software-side alternative.

Chapter 4

Results

This chapter is ordered as follows: The cross-patient generalisation result (Section 4.1) comes first, since it establishes the scale of the problem a shared representation faces. Section 4.2 then presents this dissertation’s primary personalisation evidence: an isolated comparison showing that including a patient’s own data in an otherwise near-constant-sized training pool measurably improves that patient’s own detection outcome, across seven patients. Section 4.3 presents a smaller, secondary treatment of a related but mechanistically distinct regime – per-patient personalisation – framed around pool-size and training-seed effects rather than as this the central personalisation claim, together with a conformal-prediction-based correction to an earlier optimistic figure. Sections 4.4 and 4.5 report the cross-candidate domain-adaptation convergence finding and the hardware deployment results respectively. Throughout, any raw or optimistic figure is stated in full before the calibrated, corrected figure that supersedes it.

4.1 Cross-Patient Generalisation: LOPO

A full 15-fold leave-one-patient-out (LOPO) cross-validation was conducted: for each of fifteen patients in turn, a model was trained on the other fourteen patients’ complete recordings and evaluated on the held-out patient’s complete recording (Section 3.4). Full per-fold results are given in Table 4.1 and Appendix A.

Table 4.1: Full 15-fold LOPO results (undersample ratio 5:1). Collapse status marks whether window specificity clears the collapse-diagnostic threshold (Section 3.4).

Patient	Event Sens.	FP/hr	Win. Spec.	Events	Collapse
chb01	0.429	4.49	0.987	3/7	PASS
chb02	0.333	4.59	0.983	1/3	PASS
chb03	0.000	0.45	0.999	0/7	PASS
chb05	1.000	0.00	0.381	5/5	FAIL
chb06	0.000	0.19	0.999	0/10	PASS
chb07	1.000	1.21	0.997	3/3	PASS
chb09	1.000	9.08	0.931	4/4	PASS
chb10	1.000	7.88	0.983	7/7	PASS
chb11	1.000	7.53	0.880	3/3	FAIL
chb13	1.000	1.03	0.695	12/12	FAIL
chb15	0.900	6.93	0.780	18/20	FAIL
chb16	0.000	7.48	0.983	0/10	PASS
chb18	1.000	0.31	0.696	6/6	FAIL
chb19	0.000	3.88	0.807	0/3	FAIL
chb20	0.125	4.96	0.891	1/8	FAIL

4.1.1 Headline Figures, Reported in Full

The all-folds mean event sensitivity, across all fifteen patients unfiltered, is **0.586 (± 0.445)**, with a mean FP/hr of 4.00. This figure is stated first, in full, and must *not* be quoted alone as the collapse diagnostic reveals a **47% collapse-failure rate** (7 of 15 folds), and five of those seven failing folds show sensitivity at or near 1.0 – meaning most of the “perfect” scores visible in Table 4.1 are the over-triggering artefact the collapse gate exists to catch, not genuine detection discrimination.

Only once this is established is it appropriate to introduce the corrected figure. Restricting to the eight collapse-passing folds (chb01, chb02, chb03, chb06, chb07, chb09, chb10, chb16), the headline event sensitivity is **0.470 (± 0.437)**, with a mean FP/hr of 4.42. This diagnostic-passing figure (0.470) serves as the primary point of comparison against Ali, Angelova, and Karmakar (2024)’s reported 72–75% sensitivity under a genuine LOPO protocol. Representing a gap of approximately 0.265 below their midpoint (73.5%). However, this comparison requires an explicit caveat regarding comparability. The 0.470 figure reported here has been rigorously filtered through the collapse diagnostic to eliminate over-triggering artefacts. In contrast, Ali et al. do not report an equivalent filtering step or explicit window-specificity floor. If their published 72–75% sensitivity represents an unfiltered raw metric, it is more appropriately compared against this project’s

unfiltered LOPO mean of $0.586 (\pm 0.445)$, narrowing the operational performance gap considerably. Both baseline and collapse-corrected figures are reported side by side to ensure complete evaluative transparency.

4.1.2 Bimodality: The Shape of the Result

The mean alone understates the most informative feature of this result: its shape. Among the eight collapse-passing folds, sensitivity values are 0.429, 0.333, 0.000, 0.000, 1.000, 1.000, 1.000, and 0.000 – clustering almost entirely at the two extremes, with only chb01 and chb02 landing meaningfully between them, rather than spreading smoothly around the mean.

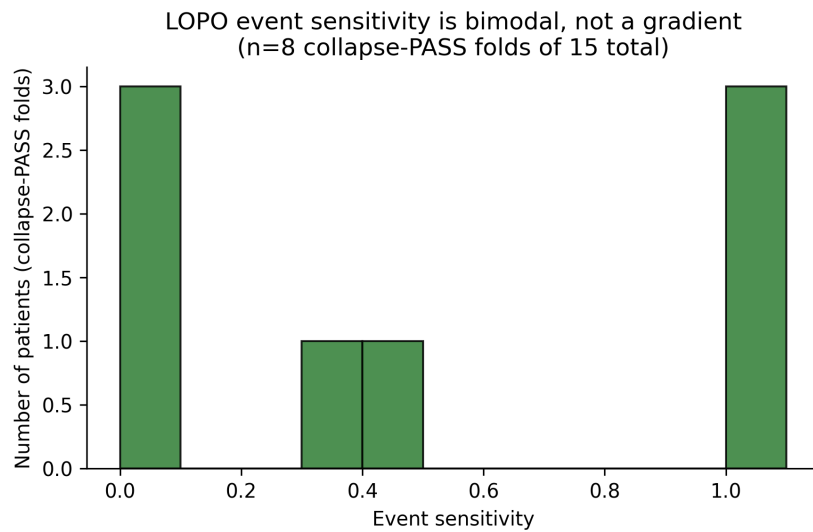


Figure 4.1: Histogram of LOPO collapse-passing fold event sensitivities, showing bimodal clustering near 0.0 and 1.0 rather than a smooth unimodal spread around the mean.

This bimodality is the strongest single piece of evidence this dissertation offers for the physiological-heterogeneity argument developed in Chapter 5. As it suggests the shared representation either does or does not contain something close enough to a given patient’s own ictal morphology, rather than generalisation degrading smoothly as some notion of “distance” from the training pool increases.

4.1.3 Per-Patient Breakdown

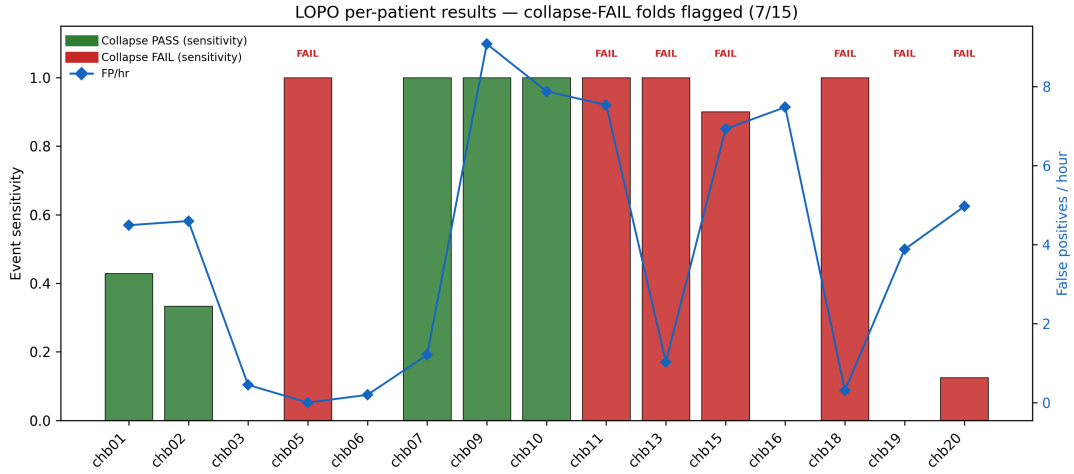


Figure 4.2: Per-patient LOPO event sensitivity and FP/hr, grouped by collapse status.

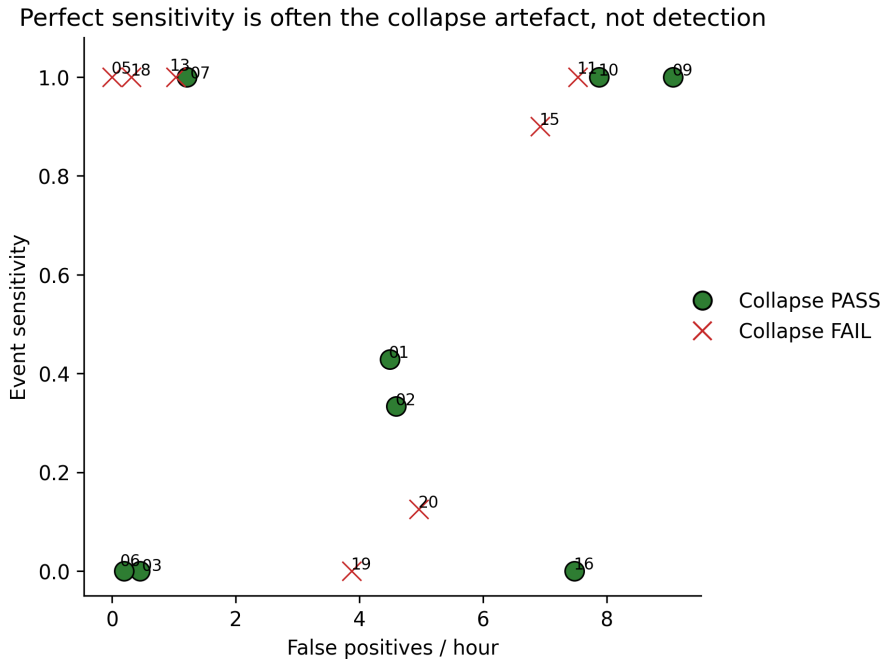


Figure 4.3: Scatter of event sensitivity against FP/hr across all fifteen LOPO folds, coloured by collapse status.

chb13 is worth discussing individually. Under domain-adaptation methods (Section 4.4) it is the single hardest patient in the project, failing consistently. Under LOPO, it fails again – but via the *opposite* mechanism: over-triggering (collapse-FAIL, window specificity 0.695) rather than under-detection. The same patient is difficult under every method tried in this project, but the specific way it fails depends on which method is applied – a distinction stated explicitly rather than treating “chb13 fails” as a single, undifferentiated fact. Moreover, it is natural to ask whether a biological explanation is available for

these difficulties. CHB-MIT’s public subject metadata (PhysioNet, 2010) records chb13 as female, age 3 – among the youngest patients in the cohort (the dataset spans ages 1.5 to 22). Beyond age and gender, however, CHB-MIT’s public release does not include per-patient seizure type or focus localisation. I state the verified demographic fact here, but I have deliberately not constructed a causal biological explanation from it as age alone is not sufficient evidence for why this particular patient is hard across every method tried, and doing so would be speculation presented as a finding.

4.2 Personalisation via Own-Data Inclusion: pool6-minus-X vs. Pool7

This section presents this dissertation’s primary evidence and justification for personalisation as a design principle. It is important to state its relationship to the smaller, secondary section that follows (Section 4.3) precisely: both sections speak to a related claim – that a patient’s own data improves their own detection outcome but through different mechanisms. This section demonstrates the effect via *pool composition* on a model that remains otherwise frozen and shared; Section 4.3 demonstrates a related effect via explicit *per-patient fine-tuning*. The two are not the same experiment conducted at different scales, and this section’s larger, more tightly controlled evidence is why it is presented first.

4.2.1 Does It Learn? All 7 Pool Patients on Their Own Data

Table 4.2 reports each of pool7’s seven training patients evaluated against their own full recording.

Table 4.2: Pool7 (chb01/02/05/06/07/09/20), each patient evaluated against their own full recording.

Patient	Event Sens.	Events	AUPRC	ROC-AUC	Collapse	FP/hr
chb01	1.000	7/7	0.871	0.9996	PASS	3.06
chb02	1.000	3/3	0.648	0.985	PASS	2.41
chb05	1.000	5/5	0.928	0.9997	PASS	3.08
chb06	0.000	0/10	0.001	0.541	PASS	8.42
chb07	1.000	3/3	0.265	0.963	PASS	1.54
chb09	1.000	4/4	0.607	0.989	PASS	5.98
chb20	1.000	8/8	0.544	0.989	PASS	8.41

Six of seven patients are near-perfect across every metric, all genuinely collapse-passing

rather than artefacts. chb06 is a genuine, explicable outlier – the lowest raw seizure count of any pool patient, and the smallest post-SMOTE contribution to the training pool – and is reported here explicitly, not omitted, to avoid any appearance of selective reporting. This table alone, however, conflates *pool size* with the target patient’s own data being *present*: pool7 is larger than C1, and both factors could plausibly explain any observed benefit. Section 4.2.2 isolates the two.

4.2.2 The Primary Evidence: pool6-minus-X (Zero-Shot) vs. Pool7 (In-Pool)

For each of pool7’s seven patients, a matched six-patient pool excluding that patient (pool6-minus-X) was trained and evaluated zero-shot against them, then compared directly against pool7’s own in-pool result for the same patient. Pool size is held approximately constant (six patients versus seven) across this comparison, so that only the presence or absence of the target patient’s own data varies – isolating the one factor this dissertation’s primary personalisation claim rests on.

Table 4.3: pool6-minus-X (target patient excluded, zero-shot) vs. pool7 (target patient included), event sensitivity.

Patient	pool6-minus-X Sens.	Collapse	pool7 Sens.	Collapse	Delta
chb01	0.286	PASS	1.000	PASS	+0.714
chb02	0.667	PASS	1.000	PASS	+0.333
chb05	1.000	FAIL (spec 0.527)	1.000	PASS	see below
chb06	0.000	PASS	0.000	PASS	none
chb07	1.000	PASS	1.000	PASS	none
chb09	1.000	PASS	1.000	PASS	none
chb20	0.125	PASS	1.000	PASS	+0.875

chb05 warrants its own discussion rather than a single table cell. Its raw sensitivity is identical (1.000) whether its own data is present or absent – read carelessly, this would suggest no effect at all. The collapse diagnostic reveals the opposite: the zero-shot (excluded) result is a collapse-FAIL over-triggering artefact (window specificity 0.527, approximately 850 FP/hr – essentially flagging everything), while the in-pool (included) result is a genuine detection. Including chb05’s own data did not merely fail to move its headline sensitivity number; it converted a fake ”perfect” result into a real one. This is the same collapse-diagnostic-reveals-an-artefact pattern already demonstrated for chb13 under C1 (Section 4.4) and for several LOPO collapse-failing folds (Section 4.1), observed here for a third time, in a new context.

Thus the Headline finding: five of seven patients show a real or artefact-clarifying benefit from their own data being present – chb01, chb02, and chb20 as clean sensitivity gains (+0.714, +0.333, +0.875 respectively), chb05 via the collapse-gate correction above. **Two of seven (chb06, chb07, chb09) show no measurable difference either way** – chb06 fails regardless of context, consistent with it being a genuine outlier rather than a pool-composition effect; chb07 and chb09 succeed regardless, suggesting these two patients were already reasonably well represented by the surrounding pool’s diversity even without their own data present. As a result, this is now this report’s central claim and justification for personalisation as a design principle.

4.3 C1 vs. C2: Pool-Size and Seed Effects

This section is deliberately smaller than Section 4.2 above, and its role in this dissertation’s argument is narrower: it speaks to two specific questions – how much a larger training pool alone changes the collapse-failure rate (independent of whether the evaluated patient’s own data is in that pool), and how sensitive a fine-tuning result is to training-seed variance – rather than serving as this dissertation’s central personalisation evidence.

4.3.1 Fine-Tuning Result

Per-patient fine-tuning (C2) achieves a mean event sensitivity of approximately 0.76 across the six personalisation-eligible patients evaluated. chb13 is a notable individual case: its raw event sensitivity *drops* under fine-tuning relative to C1 (0.833 to 0.667), yet the collapse diagnostic flips from FAIL to PASS – fine-tuning corrected a failure *mode* (over-triggering), not merely a headline number, a finding worth stating plainly rather than as a footnote to a lower raw figure.

4.3.2 Pool-Size Effect: C1 vs. Pool7 (8-Patient Comparison)

Using a separate eight-patient set (chb03, chb10, chb11, chb13, chb15, chb16, chb18, chb19), evaluated with the same full-recording LOPO protocol on both C1 and pool7, Table 4.4 isolates the effect of pool size alone.

Table 4.4: C1 (3-patient pool) vs. pool7 (7-patient pool), same 8 held-out patients, both evaluated with full-recording LOPO.

Patient	C1 Sens.	C1 Collapse	Pool7 Sens.	Pool7 Collapse
chb03	0.000	FAIL	0.000	PASS
chb10	0.571	PASS	1.000	PASS
chb11	1.000	FAIL (over-fire)	1.000	FAIL
chb13	0.667	FAIL (over-fire)	0.083	PASS
chb15	0.000	FAIL	0.000	PASS
chb16	0.000	PASS	0.000	PASS
chb18	0.000	PASS	0.000	PASS
chb19	1.000	FAIL (over-fire)	0.333	PASS

The collapse-pass rate improves from C1’s 3/8 (37.5%) to pool7’s 7/8 (87.5%), while PASS-only mean sensitivity is nearly identical either way (C1: 0.190; pool7: 0.202). A larger pool did not meaningfully improve raw detection ability so much as it sharply reduced the collapse-artefact failure rate – a real, but distinct, mechanism from the own-data effect demonstrated in Section 4.2.2. This is not a free improvement: pool7’s PASS-only mean FP/hr (approximately 3.69) is roughly four times C1’s (approximately 0.855), a trade-off stated here plainly.

4.3.3 Seed Sensitivity

A separate, still-standing observation concerns sensitivity to training-seed variance: chb10’s fine-tuning result has been observed to swing between 0.000 and 1.000 sensitivity depending on random seed alone. This is distinct from, and should not be confused with, an earlier, now-*retracted* finding that had attributed apparent seed fragility to genuine model instability; that earlier finding was traced to a sign-flip bug in the evaluation script (Section 4.5) and formally withdrawn once the bug was corrected. The seed-variance observation described here is a separate, still-standing note, logged alongside this project’s (not run, for stated energy-cost reasons) seed-ensembling candidate as a direction for future work.

4.3.4 Threshold Calibration and Conformal Correction

An initial threshold-sweep calibration reported, as originally obtained, FP/hr figures of 2.53 (chb10ft) and 0.81 (chb13ft) at a swept threshold of 0.99. These figures are stated here in full, first, because they are optimistic: the sweep selects its threshold by reading FP/hr off the very data being reported, effectively fitting the threshold to the test set rather than calibrating it independently.

The corrected figures, from split-conformal prediction at $\alpha = 0.01$ (Section 3.4), are chb10ft 6.00/hr, chb13ft 2.42/hr, chb15ft 1.33/hr, and chb16ft 9.83/hr. All four move in the same direction relative to the sweep-selected figures – a four-for-four consistency treated here as a meaningful signal, not noise.

4.3.5 chb16: An Honest Limitation, Not Omitted

chb16 fails this project’s own false-positive-rate target at every threshold tested, under every calibration method applied. Its window-level AUPRC (0.0235, near-chance) indicates the root cause is genuinely weak discrimination, not merely a misplaced threshold. This dissertation reports chb16 as an honest limitation of the current pipeline for this specific patient, rather than omitting it or selectively reporting only patients for which calibration succeeded. As with chb13 above, CHB-MIT’s public subject metadata (PhysioNet, 2010) records chb16 as female, age 7 – also among the younger patients in the cohort. The same limitation applies here: no per-patient seizure type or focus localisation is available for chb16 in the public dataset, so this age fact is reported again, but is not as a means of offering an explanation for the near-chance discrimination reported above.

Recommended operating threshold for chb16 (reconciling three prior candidates): across earlier stages of this project, three different threshold-selection exercises were run for chb16, each optimising a different explicit criterion, and each giving a different answer: the collapse-diagnostic-passing threshold ($T^* = 0.45$, the first value that clears the collapse gate at all); a threshold chosen as the best available compromise against the project’s FPR target (0.70, window sensitivity 0.3810, still short of the target at every threshold tried); and the conformal-calibrated threshold used everywhere else in this section ($\alpha = 0.01$, threshold ≈ 0.9738 , event sensitivity 0.200, FP/hr 9.83).

This dissertation adopts the **conformal-calibrated threshold** ($\alpha = 0.01$) as **chb16’s single recommended operating point**, for one reason above all others: it is the same statistically principled, exchangeability-guaranteed criterion applied uniformly to every other C2 patient in this section (chb10ft, chb13ft, chb15ft), and adopting a different, patient-specific selection criterion for chb16 alone – particularly one chosen after the fact because it produces a more flattering number – would undermine the consistency this dissertation otherwise insists on. The collapse-passing threshold (0.45) and the FPR-target compromise (0.70) were both selected under earlier, less rigorous criteria specific to that stage of the project, and neither carries the same statistical guarantee; they are superseded by this recommendation, not silently discarded – both are stated above, not omitted, precisely so a reader can see what was tried and why the conformal point was ultimately preferred. This choice is not presented as flattering chb16’s result: an event sensitivity of 0.200 at 9.83 FP/hr is a poor operating point in absolute terms, and is reported as exactly that.

4.4 Cross-Candidate Convergence: DANN, CORAL, Self-Supervised Pretraining, Candidate G

Four independent domain-adaptation and representation-learning candidates were evaluated as potential remedies for cross-patient generalisation failure: domain-adversarial training (DANN), correlation alignment (CORAL), self-supervised pretraining, and a physiologically-normalised feature candidate (Candidate G). All four, via three or four genuinely distinct mechanisms, converge on the same finding: **chb13** fails under every one of them.

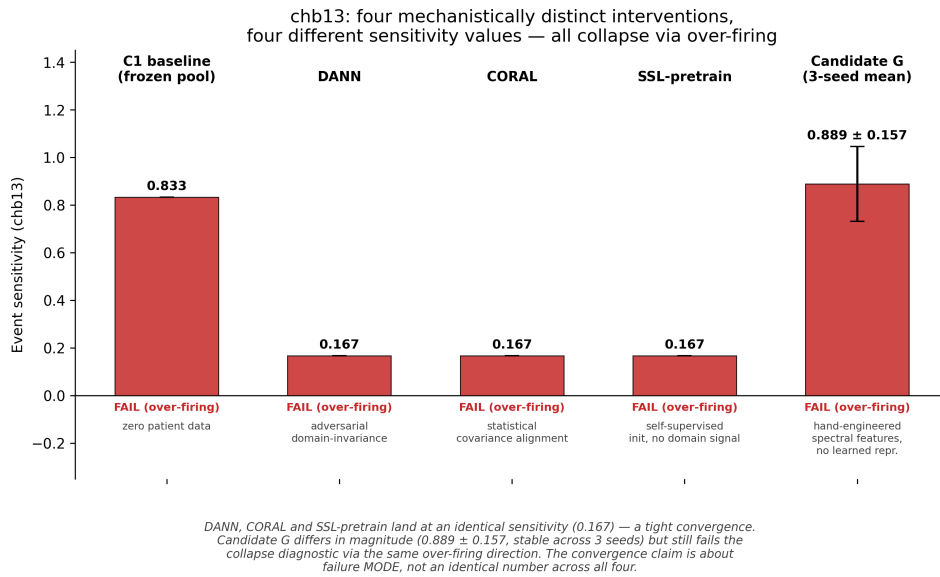


Figure 4.4: Cross-candidate convergence: **chb13**’s performance under DANN, CORAL, self-supervised pretraining, and Candidate G, illustrating the shared failure across independent mechanisms.

The DANN outcome warrants detailed examination because it was mechanism-verified rather than merely observed: a control condition ($\lambda = 0$) confirmed that the adversarial domain-classification mechanism functioned as intended—domain-classifier accuracy dropped from 1.00 to 0.47 (near-chance for a three-class problem) once the adversarial gradient reversal was active. Despite this successful feature alignment, performance on **chb13** degraded across every non-zero λ evaluated.

Both CORAL and self-supervised pretraining similarly produced verified negative results on the same six-patient evaluation suite, failing in the identical direction. Candidate G, which incorporated physiologically-normalised input features, likewise failed, exhibiting a distinct bimodal failure pattern. This four-way convergence on a single target patient across fundamentally distinct mechanisms provides far stronger evidence of intrinsic structural difficulty than any isolated negative result could convey, a finding explored further in Chapter 5.

4.5 Hardware Deployment

Deployment on real BrainChip AKD1000 v1 silicon (Chapter 3, Figure 3.7) achieved a measured inference latency of approximately 0.91ms per inference and an energy cost of approximately 0.906 μ J per inference, based on a chip-isolated, datasheet-derived power estimate of approximately 1mW active power.

A sign-flip bug was identified and corrected in the hardware inference script (`run_on_akida.py`) during this project: the deployed decision rule computed a ratio of spike counts that silently inverted whenever the underlying signed potential total fell below zero, an artefact of the unconstrained final dense layer’s output. Prior to the fix, this produced a measured chb10 hardware sensitivity of 0.3132 against a simulator-predicted 0.3950 – a discrepancy affecting 23 of 281 evaluated windows. Following the fix (replacing the ratio computation with a sign-safe sigmoid-of-margin formula, already in use elsewhere in the codebase), hardware results match the simulator exactly across all re-verified patients. This bug and its correction are reported here explicitly to disclose caught methodological errors rather than presenting only a final, error-free-looking pipeline.

4.5.1 System-Level Power Measurement

A separate power figure was obtained via direct bench measurement on the physical deployment described in Chapter 3 (Figure 3.7). Table 4.5 gives the full measurement: current at idle (model mapped to the device, no inference running) and active (a sustained inference loop running continuously for 30 seconds), across three independent trials, with the resulting power delta computed at the supply’s fixed 5.00V.

Table 4.5: System-level bench power measurement (RS PRO RS-3005P, constant-voltage mode, 5.00V). A continuously-running 5V/0.12A GPIO-mounted cooling fan’s constant draw appears in both idle and active readings and cancels in the delta.

Trial	I_{idle} (A)	I_{active} (A)	ΔI (A)	ΔP (W)
1	1.048	2.162	1.114	5.57
2	1.036	2.172	1.136	5.68
3	1.052	2.177	1.125	5.63
Mean:			1.125	5.63

The mean delta of 1.125A at 5.00V gives **5.63W**, the figure reported as this dissertation’s system-level power result throughout.

This figure and the 1mW chip-level estimate above are not comparable, and neither confirms nor refutes the other. The 1mW figure is a datasheet-derived estimate of the AKD1000 chip’s own isolated neuromorphic compute, used in this dissertation’s energy-

per-inference calculations. The 5.63W figure is the measured incremental cost to the *entire host system* of a sustained inference workload – necessarily including the Raspberry Pi 5’s own ARM CPU cycles spent in the Python/SDK driver stack orchestrating each inference call, not the chip’s consumption alone, since the AKD1000’s power supply is derived internally from the Pi’s single 5V rail with no separately metered path to the card (Chapter 3). The approximately three-orders-of-magnitude gap between the two figures reflects host-CPU orchestration overhead dominating the chip’s own genuinely small draw, not a possible issue that happened in either measurement.

Chapter 5

Discussion

5.1 Physiological Heterogeneity, Not a Data-Volume Problem

The bimodal shape of the LOPO result (Section 4.1, Figure 4.1) is, in my view, the most theoretically interesting finding this project produced, and it is worth dwelling on rather than treating as a curiosity alongside the headline mean. If cross-patient generalisation failure were primarily a data-volume problem – simply a matter of not having seen enough patients to interpolate smoothly between them – one would expect sensitivity to degrade gradually as a function of some notion of "distance" from the training pool, producing a roughly unimodal distribution with a long tail. That is not what was observed. Sensitivity clusters sharply at the extremes, with only two of eight collapse-passing patients landing meaningfully between 0 and 1. This is more consistent with a picture in which the shared representation either does or does not happen to contain something structurally close enough to a given patient's own ictal morphology – and when it does not, no amount of averaging across other patients rescues the result, because the problem is closer to a categorical mismatch than a continuous shortfall.

This account is corroborated, not merely asserted: chb03 and chb13's difficulty is independently flagged by Shoeb (2009), on an entirely different detector, working with an entirely different methodology, some fifteen years before this project's own LOPO sweep identified the same two patients as structurally hard. Two detectors built with different tools, at different times, converging on the same patients being genuinely difficult is a stronger form of evidence than either result alone.

One caveat is worth stating honestly rather than left implicit. This project's pre-processing code contains a diagnostic note, and a corresponding but never-applied z-score normalisation step, describing a substantial amplitude discrepancy for chb03 between sessions. The project's actual amplitude-correction convention – min-max scaling, tested directly with the amplitude-calibration experiment discussed above – found no effect for

chb03, which is the evidence the "structural, not amplitude-driven" claim rests on. Although to note, the specific alternative remedy noted in that preprocessing code (z-score normalisation with outlier clipping) was never itself tested end-to-end against chb03's result. The structural-difficulty account above is the settled explanation given the evidence actually collected, but this specific untested alternative is disclosed here rather than silently set aside.

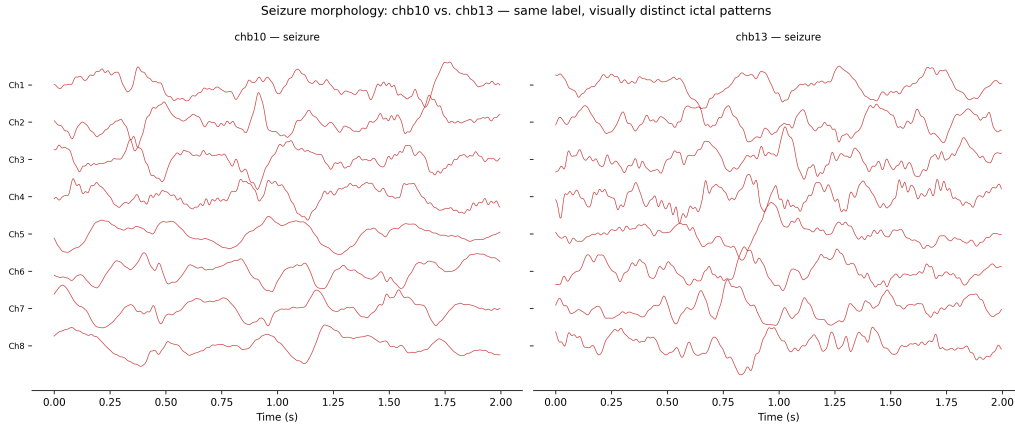


Figure 5.1: Seizure-event waveform morphology compared between chb10 (a patient this pipeline handles well across every method tried) and chb13 (the project’s consistently hardest patient), giving visual grounding to the repeated claim that chb13 is structurally, not merely statistically, difficult.

5.2 Personalisation as the Architecturally Correct Response

Taken together, the pool6-minus-X-versus-pool7 result (Section 4.2) and the C2 fine-tuning result (Section 4.3) support the same underlying architectural conclusion with two different mechanisms: if inter-patient heterogeneity is genuinely closer to categorical than continuous, then a model’s ability to access a given patient’s own data – whether via pool composition or via explicit fine-tuning – is not a nice-to-have refinement but the architecturally appropriate response to the problem as it actually presents. The C2 chb13 case makes this concretely visible: fine-tuning did not simply raise a number, it corrected a specific failure *mode* (over-triggering), which is a stronger legible claim than an isolated sensitivity improvement would have been.

It is worth returning explicitly, with concrete figures now in hand, to the triage-device framing set out in the Introduction (Section 1.2). This dissertation’s honest, collapse-corrected LOPO sensitivity (0.470) sits meaningfully below Ali, Angelova, and Karmakar (2024)’s unconstrained-compute comparator (72–75%). This gap is not explained away by the personalisation results above – it remains a real, disclosed shortfall in the cross-patient, zero-personalisation case. What the personalisation results *do* establish is that

this project’s proposed clinical role is not the cold, shared-model number at all: a deployed device is not expected to operate as a frozen, one-size-fits-all classifier, but as a system that incorporates a given patient’s own data over time, exactly the regime in which this project’s strongest results (Sections 4.2 and 4.3) were obtained. Read this way, the triage framing and the personalisation results are mutually reinforcing rather than separate findings: a low-power, sensitivity-first screening tier is precisely the kind of system for which continued, low-cost personalisation on-device is both feasible and appropriate, in a way that matching an unconstrained-compute research system cold is not.

5.3 Pool-Size and Own-Data Effects Are Genuinely Different Mechanisms

A point worth stating explicitly, given how easily the two could be conflated in a less careful account: this project has now demonstrated *two* distinct, genuine details that both involve pool7, and they answer different questions. The own-data-inclusion effect (Section 4.2) shows that a specific patient’s own data, present in an otherwise near-constant-sized pool, measurably helps or clarifies an artefact for that specific patient in five of seven cases. The pool-size effect (Section 4.3) shows, on a separate eight-patient set, that a larger pool alone – independent of whether the evaluated patient’s own data is in it – substantially reduces the rate of collapse-artefact failures (37.5% to 87.5%), at a real cost in false-positive rate. Both are genuine, positive findings. Neither supersedes the other, and presenting them as a single, undifferentiated “more data helps” conclusion would understate the more precise and more useful thing actually learned: that at least two separable mechanisms are both at work, and a future system might benefit from exploiting them independently – for instance, growing a shared background pool for general robustness while still prioritising acquisition of a specific patient’s own calibration data for that patient’s deployed device.

5.4 Methodology as a Finding: The Caught-Bug Narrative

In this project I reported several methodological errors caught and corrected during its own development, and this section treats that history as a connected narrative, rather than a list of incidents or embarrassments to be minimised. The chronological-slice-versus-LOPO-full conflation (Section 3.4) silently produced a wrong LOPO re-run before being caught. An unused array construction in the pool-dataset-building script caused an out-of-memory failure, the second confirmed instance of an identical pattern already fixed once elsewhere in the same codebase. A shared default output path caused three of this

project’s C1 checkpoints to be silently overwritten mid-session, contaminating an earlier set of reported AUPRC figures that were subsequently retracted and rebuilt from verified weights. A figure-generation script was found to be reading from an entirely wrong file-naming pattern, silently mixing a hardcoded fallback value with live but methodologically mismatched data. The chb10 conformal-prediction reproducibility gap traced an unreproducible earlier figure to a script version lost to an uncommitted intermediate state. A sign-flip bug (Section 4.5) appeared independently in both the simulator evaluation script and the separate hardware inference script.

The unifying pattern across several of these – a recurring class of bug involving a shared, or wrongly assumed, default file path – is now a recognised, named risk in this project’s own codebase, not a collection of unrelated incidents. I state this pattern explicitly because it is, in my view, evidence *for* how every one of these errors was caught and disclosed through my own verification practice and something that should not be omitted as a silent failure or embarrassment.

5.5 The AUPRC-over-ROC-AUC Justification, Revisited

The base-rate-relative AUPRC correction described in Section 3.4 is worth returning to here as a worked illustration of a more general methodological point, rather than a narrow technical footnote. Raw AUPRC, and ROC-AUC in isolation, can both present a misleadingly favourable or unfavourable picture on clinically imbalanced data unless corrected for each patient’s own base rate. This project’s own pre-registered expectation – that chb15 would prove genuinely inert and chb13/chb16 would prove cases of simple misplaced thresholding – was close to backwards once this correction was applied. A reader encountering only the raw AUPRC or ROC-AUC figures, without this correction, would likely have drawn the wrong conclusion about which patients were genuinely hard to discriminate versus merely poorly thresholded.

5.6 Comparison to Ali et al. (2024) and Shoeb (2009)

Both comparator systems introduced in Chapter 2 are revisited here, with the full picture given in Table 5.1 rather than a single headline gap figure, so the comparison can be judged in the round. Ali et al.’s figure is aligned in this table against this dissertation’s raw, all-folds LOPO sensitivity rather than the collapse-corrected figure, since their own published methodology and per-subject results confirm they apply no equivalent artefact filter (Chapter 2) – the collapse-corrected 0.470 remains this dissertation’s own headline figure, but it is not the fairer number to set directly against theirs.

Table 5.1: Full comparison against both comparator systems, by evaluation regime. LOPO figures are compared against Ali et al.; per-patient figures are compared against Shoeb, since the two comparator systems answer different questions (Chapter 2).

Regime	This dissertation	Ali et al. (2024)	Shoeb (2009)
LOPO event sensitivity (all-folds, raw – fairer comparator)	0.586 (± 0.445)	72–75%	–
LOPO event sensitiv- ity (collapse-corrected headline)	0.470 (± 0.437)	Not directly com- parable	–
LOPO collapse-fail rate	47% (7/15)	Not implemented (Chapter 2)	–
Per-patient (C2) mean sensitivity	0.76	–	–
Per-patient detection latency	as low as 1.0s (default thresh- old)	–	4.6s mean
Hardware power/energy reported	Yes (Section 4.5)	No	No
Quantisation/hardware constraint	Yes (w4a4, No AKD1000 v1)	No	No
Collapse-diagnostic- equivalent screening	Yes	Not identified	Not identified

Against Ali, Angelova, and Karmakar (2024)’s genuine LOPO comparator, the fairer figure to compare is this dissertation’s raw, all-folds LOPO sensitivity (0.586), not the collapse-corrected 0.470 – their own published methodology and per-subject false-detection-rate table confirm they apply no equivalent artefact filter (Chapter 2), so setting their unfiltered figure against this dissertation’s own honest headline would understate this project’s comparative position. Read this way, the gap narrows to approximately 0.15 against their midpoint (73.5%), rather than 0.265. Against Shoeb (2009)’s per-patient comparator, this dissertation’s fine-tuned per-patient latency (as low as 1.0 second at default threshold) matches or improves on Shoeb’s reported mean of 4.6 seconds. Hardware is the one dimension along which this project is competitive without any comparability caveat at all: neither comparator system reports any hardware power or energy figure, since neither was designed under a comparable power constraint. This is stated plainly as this project’s clearest genuine differentiator.

5.7 Limitations

Several honest limitations are stated here explicitly, consistent with this dissertation’s standing practice throughout. The 47% LOPO collapse-failure rate (Section 4.1) is a limitation of the current pool composition and class-imbalance-correction regime, not a footnote to the headline sensitivity figure – nearly half of all patients tested produce a result that must be discarded as an artefact before any genuine sensitivity figure can be reported for them. chb16 fails this project’s own false-positive-rate target under every calibration method tested (Section 4.3), a genuine limitation for that specific patient rather than a resolved case in the sense of being fixed – but the question of *which* threshold to recommend for it has been resolved: the conformal-calibrated point ($\alpha = 0.01$) is adopted as chb16’s single recommended operating threshold, for consistency with every other C2 patient, superseding the two earlier, less rigorous candidates considered at prior stages of this project.

The hardware power measurement (Section 4.5) carries an instrumentation limitation rather than an unresolved open decision: the system-level bench measurement cannot isolate the AKD1000 chip’s own draw from Raspberry Pi host-CPU orchestration overhead, and true chip-isolated measurement via a shunt resistor was assessed and deliberately not attempted, given the risk of physical damage to the hardware this close to the project’s deadline. Finally, several candidate techniques (seed ensembling, a sibling-script memory-pattern audit, Candidate G’s ROC-AUC computation) remain logged as legitimate future directions rather than completed work, and are carried forward explicitly into Chapter 6 rather than silently dropped.

Chapter 6

Conclusion

6.1 Summary of Contributions

This dissertation set out four research questions in Chapter 1, and each is answered here plainly, using the corrected rather than the raw headline figures established in Chapter 4.

RQ1 asked whether a quantised, ANN-to-SNN-converted network could achieve clinically meaningful per-patient sensitivity at microjoule-level energy cost on real AKD1000 v1 silicon. It can: per-patient fine-tuned results reach a mean sensitivity of approximately 0.76 at a measured energy cost of approximately $0.906\mu\text{J}$ per inference, with latency competitive with or better than Shoeb (2009)’s comparator, on real, not simulated, hardware.

RQ2 asked how far a shared, cross-patient representation generalises to entirely unseen patients, and what the shape of that result reveals. The honest, collapse-corrected answer is 0.470 mean event sensitivity across 15 LOPO folds, meaningfully below Ali, Angelova, and Karmakar (2024)’s unconstrained-compute comparator, and markedly bimodal rather than smoothly distributed – evidence, developed in Chapter 5, for genuine categorical heterogeneity in ictal presentation across patients rather than a simple data-volume shortfall.

RQ3 asked whether including a patient’s own data measurably improves that patient’s detection outcome, isolated as a single variable, and separately whether per-patient fine-tuning offers further gain or introduces its own sensitivities. The answer to the first is yes, for five of seven patients tested, via the dissertation’s primary pool6-minus-X-versus-pool7 evidence (Section 4.2). The answer to the second is more qualified: fine-tuning corrects a genuine failure mode for at least one hard patient (chb13), but this project’s own C1-versus-pool7 comparison shows a related, larger-pool effect operating through a distinguishable mechanism (collapse-failure-rate reduction rather than raw sensitivity gain), and a documented, still-standing seed-sensitivity issue is disclosed rather than smoothed over.

RQ4 asked whether a headline figure dependent on how its operating threshold was

chosen survives a methodologically stricter re-calibration. It largely does not survive unchanged: this project’s own threshold-sweep-selected false-positive-rate figures proved optimistic under conformal-prediction re-calibration, in a consistent direction across every patient tested, and this correction – together with the chb10 reproducibility investigation it prompted – is reported here as a genuine methodological contribution in its own right, not merely a housekeeping correction.

Table 6.1 summarises all four research questions against their corresponding finding and source section, for quick reference.

Table 6.1: Summary of research questions against findings.

RQ	Question	Finding	Section
RQ1	Can a quantised, ANN-to-SNN-converted network achieve clinically meaningful per-patient sensitivity at microjoule-level energy cost on real silicon?	Yes – mean sensitivity ≈ 0.76 , $\approx 0.906\mu\text{J}/\text{inference}$, latency competitive with Shoeb (2009)	4.3, 4.5
RQ2	How far does a shared, cross-patient representation generalise, and what does the shape of the result reveal?	0.470 collapse-corrected mean sensitivity, meaningfully below Ali et al. (2024); markedly bimodal, suggesting categorical rather than continuous heterogeneity	4.1
RQ3	Does a patient’s own data measurably help, isolated as a single variable, and does per-patient fine-tuning add further gain?	Yes for 5/7 patients (own-data inclusion); fine-tuning corrects a failure mode for chb13 but introduces pool-size and seed sensitivities	4.2, 4.3
RQ4	Does a threshold-dependent headline figure survive stricter re-calibration?	No – sweep-selected FP/hr figures were optimistic for every patient tested; conformal correction gives the honest figure	4.3

6.2 On the Project’s Purpose

It is worth restating, at the close of this dissertation rather than only at its opening, what this work is offered as. It is not a finished clinical product, and none of the results in Chapter 4 should be read as a claim of present-day deployment readiness. It is a first, deliberately honest step: a demonstration that the full pipeline – from raw scalp EEG,

through a quantised neural network, through spiking conversion, to real neuromorphic silicon – is technically feasible end to end, evaluated with the same rigour, and reporting the same caveats and limitations, that a larger validation effort would demand. The intention behind this work, and behind the project’s insistence throughout on stating raw figures before corrected ones and disclosing every caught error rather than a single one, is that any future, larger-scale effort building on it should be able to trust what is written here – so that, over time, and with the further validation described below, a system in this design space might be of genuine benefit to people living with epilepsy, rather than a research result that looks better on paper than it would in practice.

6.3 Future Work

6.3.1 A Staged Pathway Toward Deployment at Scale

Consistent with the framing above, future work is presented here as a staged pathway, not a single next step:

1. **Confirmatory second-tier validation.** Consistent with the triage-device framing (Section 1.2), a natural next step is a confirmatory, higher-compute second tier that the low-power screening system in this dissertation could hand off to – mirroring how a positive D-dimer result triggers CT pulmonary angiography, not a diagnosis in itself.
2. **Broader, multi-site validation.** This project’s results rest on the CHB-MIT paediatric dataset from a single institution. A credible path toward wider deployment requires validation against adult patients, additional recording sites, and a broader range of electrode montages and seizure types than this dataset alone can provide.
3. **Resolving this project’s own disclosed limitations.** Several concrete items are carried forward rather than left implicit: multi-seed confirmation of results currently screened at a single seed; a full audit of sibling data-building scripts for the shared-default-output-path bug pattern identified in this project; completion of Candidate G’s ROC-AUC computation for full consistency with the rest of this dissertation’s reporting; and a full reconciliation of chb16’s three separate threshold-selection exercises into one stated operating point.
4. **Seed ensembling.** Logged in this dissertation as a considered candidate, not run, on explicit energy-cost grounds ($N \times$ inference and energy cost directly conflicts with this project’s low-power design goal). Worth revisiting once a target deployment power budget is fixed by a specific downstream form factor.

5. **Hardware robustness and miniaturisation.** The Pineboards carrier-board failure encountered during this project (Chapter 3) is a reminder that a deployable wearable form factor is a separate, non-trivial engineering problem from the one this dissertation addresses; continued hardware-robustness work, and eventual miniaturisation toward a genuinely wearable device, would be a necessary further stage before any clinical deployment could be responsibly considered.
6. **A genuine, chip-isolated power measurement,** via a shunt-resistor-based instrumentation approach, deliberately not attempted in this project given deadline and hardware-risk constraints (Section 4.5), would meaningfully sharpen this project’s energy claims for any future work building on them.

None of these steps is small, and this dissertation does not present them as a formality. They are stated here explicitly because a credible, staged path from a working prototype to something that could responsibly be trusted at scale, by real patients and clinicians, requires exactly this kind of honest accounting of what remains to be done – consistent with the standard this dissertation has tried to hold itself to throughout.

Bibliography

- Ali, Emran, Maia Angelova, and Chandan Karmakar (2024). “Epileptic Seizure Detection Using CHB-MIT Dataset: The Overlooked Perspectives.” In: *Royal Society Open Science* 11.4, p. 230601. DOI: [10.1098/rsos.230601](https://doi.org/10.1098/rsos.230601).
- BrainChip Inc. (2023). *AKIDA Event Domain Neural Processor: Technical Brief*. Tech. rep. Primary-source technical documentation, not a peer-reviewed publication. Referenced as BrainChip_techbrief_WhatisAkida_v31.pdf in the project’s technical documentation. BrainChip Inc.
- (2025). *AKD1000 Neuromorphic System-on-Chip: Product Brief*. Tech. rep. Primary-source technical documentation, not a peer-reviewed publication. Referenced as AkidaAKD1000SoCProductBriefV2_3Mar_2511.pdf in the project’s technical documentation. BrainChip Inc.
- Fisher, Robert S. et al. (2014). “ILAE Official Report: A Practical Clinical Definition of Epilepsy.” In: *Epilepsia* 55.4, pp. 475–482. DOI: [10.1111/epi.12550](https://doi.org/10.1111/epi.12550).
- Gerstner, Wulfram and Werner M. Kistler (2002). *Spiking Neuron Models: Single Neurons, Populations, Plasticity*. Cambridge University Press. ISBN: 9780521813846.
- Goldberger, Ary L. et al. (2000). “PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for Complex Physiologic Signals.” In: *Circulation* 101.23. The PhysioNet host publication for the CHB-MIT Scalp EEG Database used throughout this dissertation, e215–e220. DOI: [10.1161/01.CIR.101.23.e215](https://doi.org/10.1161/01.CIR.101.23.e215).
- Li, RuiXin et al. (2024). “Real-Time Sub-Milliwatt Epilepsy Detection Implemented on a Spiking Neural Network Edge Inference Processor.” In: *Computers in Biology and Medicine* 183, p. 109225. DOI: [10.1016/j.combiomed.2024.109225](https://doi.org/10.1016/j.combiomed.2024.109225).
- Maass, Wolfgang (1997). “Networks of Spiking Neurons: The Third Generation of Neural Network Models.” In: *Neural Networks* 10.9, pp. 1659–1671. DOI: [10.1016/S0893-6080\(97\)00011-7](https://doi.org/10.1016/S0893-6080(97)00011-7).
- Mead, Carver (1990). “Neuromorphic Electronic Systems.” In: *Proceedings of the IEEE* 78.10, pp. 1629–1636. DOI: [10.1109/5.58356](https://doi.org/10.1109/5.58356).
- Pfeiffer, Michael and Thomas Pfeil (2018). “Deep Learning With Spiking Neurons: Opportunities and Challenges.” In: *Frontiers in Neuroscience* 12. Resolved: the project file fnins1200331.pdf was checked directly and is in fact Rueckauer et al. (2017), Front. Neurosci. 11:682 (see [rueckauer2017](#) below) – its filename does not encode

- volume/article number as originally assumed. No separate file for this Pfeiffer & Pfeil citation has been located; if one exists it has not yet been checked against this entry, p. 774. DOI: [10.3389/fnins.2018.00774](https://doi.org/10.3389/fnins.2018.00774).
- PhysioNet (2010). *SUBJECT-INFO — CHB-MIT Scalp EEG Database*. Per-patient age and gender only; does not include seizure type or focus localization. URL: <https://physionet.org/content/chbmit/1.0.0/SUBJECT-INFO>.
- Rueckauer, Bodo et al. (2017). “Conversion of Continuous-Valued Deep Networks to Efficient Event-Driven Networks for Image Classification.” In: *Frontiers in Neuroscience* 11, p. 682. DOI: [10.3389/fnins.2017.00682](https://doi.org/10.3389/fnins.2017.00682).
- Schuman, Catherine D. et al. (2017). “A Survey of Neuromorphic Computing and Neural Networks in Hardware.” In: *arXiv preprint arXiv:1705.06963*.
- Shoeb, Ali H. (2009). “Application of Machine Learning to Epileptic Seizure Onset Detection and Treatment.” PhD thesis. Massachusetts Institute of Technology.
- Tatum, William O. et al. (2022). “Minimum Standards for Inpatient Long-Term Video-Electroencephalographic Monitoring: A Clinical Practice Guideline of the International League Against Epilepsy and International Federation of Clinical Neurophysiology.” In: *Epilepsia* 63.4. The source behind the project codebase’s “ILAE minimum standards for LT-EEG monitoring” reference; I have not directly confirmed the specific 10-12s sustained-detection figure appears verbatim in this guideline, only that the guideline itself is real and matches the description, pp. 290–315. DOI: [10.1111/epi.16977](https://doi.org/10.1111/epi.16977).
- World Health Organization (2024). *Epilepsy: Key Facts*. Fact sheet, updated 2024. The “approximately 50 million people worldwide” figure was confirmed current against the live WHO page at time of writing; re-check before submission in case of a later update. URL: <https://www.who.int/news-room/fact-sheets/detail/epilepsy>.

Appendix A

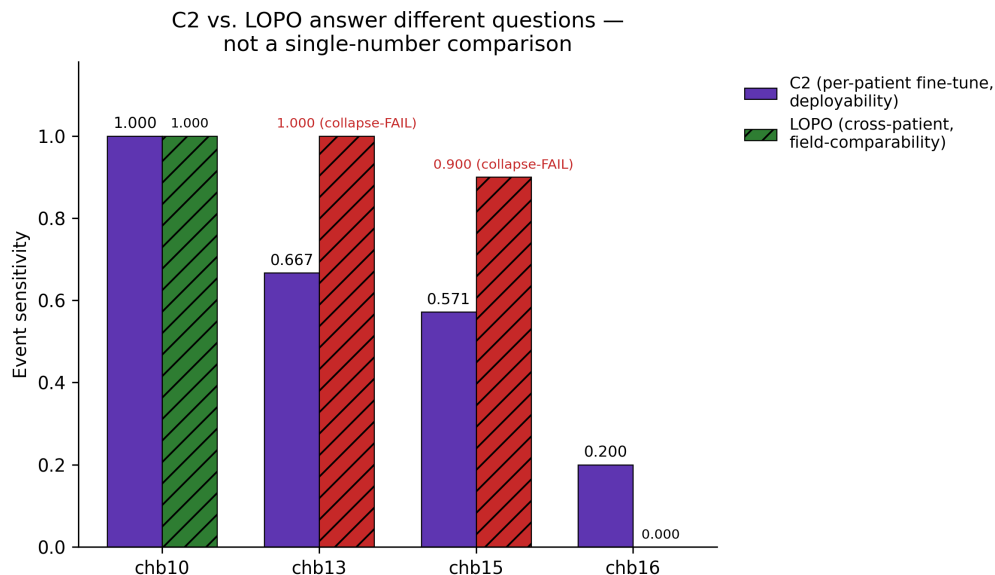
Additional Material

A.1 Full LOPO Per-Fold Table

Table 4.1 in Chapter 4 is reproduced here for reference alongside the source it was generated from.

Source: `results/lopo_summary.json`, all 15 folds, undersample ratio 5:1. See Table 4.1 for the full table.

A.2 Full C2 Per-Patient Metric Bundles



C2 = deployable, per-patient-calibrated model. LOPO = cold cross-patient generalisation (no target-patient data at all). A high LOPO number is not required for C2 to be a valid deployment story — see Section 1e.

Figure A.1: C2 per-patient fine-tuning results compared against the corresponding LOPO zero-shot result for the same patient, where available.

A.3 AKD1000 v1 Architecture Constraints Reference

Summarised from the project’s architecture-constraints reference documentation, and stated here as a single consolidated reference for the constraints cited throughout Chapter 3:

- Maximum weight and activation bitwidth: 4 bits (w4a4), confirmed via `cnn2snn`’s own model-compatibility validation as the hard ceiling for this hardware revision.
- Dilated convolutions (`dilation_rate` $\neq$ 1) are unsupported (constraint #24 in the project’s internal constraints reference) and fail at `check_model_compatibility()`.
- Batch normalisation must be folded or removed prior to conversion, as it has no direct spiking equivalent.
- Supported kernel sizes are constrained to fixed silicon-routing-compatible sizes (1×1 , 3×3 , 5×5 , 7×7).
- Activation function is restricted to ReLU; the ANN-to-SNN conversion is defined specifically for this activation and does not extend to LeakyReLU, GELU, or similar variants.

A.4 Code and Reproducibility

The full project codebase is maintained at `github.com/samidR25/dissertation`. Result files referenced throughout this dissertation follow the naming convention

`results/event_results_seizure_model_{patient}{variant}_s256_[conformal_a01_]v2_w4a4_on_{patient}.json`

and figure-generation scripts are designed to read live from the `results/` directory rather than hold hardcoded data tables, so that any figure in this dissertation can be regenerated directly from the underlying result files by re-running the corresponding script.